THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 4 OF 4

STACK SPEED & FLUENCY

60 Progressive Mathematical Speed & Fluency Practice Tests (1,200 Problems Across 15 Sections)

A self-contained mathematical speed and fluency test book for homeschooling families, classrooms, and math clubs. Every section provides four compact 20-problem practice tests of increasing difficulty—Test A (Level 1: Foundation), Test B (Level 2: Core Fluency), Test C (Level 3: Multi-Step Speed), and Test D (Level 4: Advanced Challenge)—with clean answer blanks and complete self-check answer keys.

StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Calculating Without Parentheses (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Lessons & Complete Answer Keys
- Book 2 — Hands-On Math Lab: K–12 Student Manipulative & Four-Line Stack Exploration Workbook
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method Student Practice Workbook (900 Problems) & 15-Page Coach Guide
- Book 4 — Stack Speed & Fluency: 15 Progressive Timed Sprints for Inside-Out Mathematical Fluency

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio (PLACE-01): 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio (FRAC-01): 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio (PRIME-01): Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

COPYRIGHT, STATE STANDARDS ALIGNMENT & OPEN-EDUCATION LICENSE

STACK SPEED & FLUENCY: 60 Progressive Mathematical Speed & Fluency Practice Tests (1,200 Problems Across 15 Sections)

Copyright © 2026 by StackCalc (https://stackcalc.io). Some rights reserved. Published by StackCalc Learning Lab, San Francisco, California. First Edition (US Letter 8.50 × 11.00 in / 215.9 × 279.4 mm).

State & District Curriculum Alignment:

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

Open-Education License (CC BY-NC-SA 4.0) & Perpetual Classroom Reproduction Permission:

Classroom teachers, math coaches, interventionists, homeschool families, and schools are expressly permitted to photocopy or print student lab worksheets, speed trials, reference cards, and parent welcome letters for non-commercial educational instruction under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International.

Regulatory & Physical Safety Notice (Reese's Law / P.L. 117-171 / 16 CFR Part 1263 & ASTM F963):

Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

Trademark Notice & Print Colophon:

StackCalc, SC-32, and SC-6 are trademarks of StackCalc. All curriculum and textbook names are trademarks of their respective owners, none of whom endorse or sponsor this independent publication. Typeset in Computer Modern Unicode (CMU Serif), SpaceMono, and STIX Math in high-contrast monochrome ink on archival white paper.

Table of Contents

Directory of the Preface, Master Score Tracker, 15 Progressive Test Sections (Tests A–D), and Complete Answer Keys.

Section / Progressive Practice Test Series	Practice Tests Included	Difficulty Progression	Pages
Preface: How to Use These Progressive Speed Tests	Students & Families	Untimed Mastery & Timed Speed Modes	**p. iv**
Master Score & Personal-Best Progress Tracker	All 60 Practice Tests	Score Log Across Levels 1 to 4	**p. 1**
Section 1: Step Counting, Benchmarks & Sequences	Tests 1A, 1B, 1C, 1D (80 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 2–3**
Section 2: Number Bonds & Missing Addends	Tests 2A, 2B, 2C, 2D (80 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 4–5**
Section 3: Complements & Make-10 Bridging	Tests 3A, 3B, 3C, 3D (80 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 6–7**
Section 4: Doubles, Equal Groups & Squares	Tests 4A, 4B, 4C, 4D (80 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 8–9**
Cumulative Review C1: Number Sense (Sections 1–4)	Tests C1A, C1B, C1C, C1D (80 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 10–11**
Section 5: Multi-Step Operations & Grouping	Tests 5A, 5B, 5C, 5D (80 probs)	Level 1 → Level 4 (Grades 3–8+)	**pp. 12–13**
Section 6: Fractions, Decimals & Reciprocals	Tests 6A, 6B, 6C, 6D (80 probs)	Level 1 → Level 4 (Grades 4–9+)	**pp. 14–15**
Section 7: Primes, GCF/LCM & Nested Expressions	Tests 7A, 7B, 7C, 7D (80 probs)	Level 1 → Level 4 (Grades 5–10+)	**pp. 16–17**
Section 8: Algebra, Quadratics & Coordinate Geometry	Tests 8A, 8B, 8C, 8D (80 probs)	Level 1 → Level 4 (Grades 6–10+)	**pp. 18–19**
Cumulative Review C2: Intermediate Math (Sections 1–8)	Tests C2A, C2B, C2C, C2D (80 probs)	Level 1 → Level 4 (Grades 5–10+)	**pp. 20–21**
Section 9: Trigonometry, Vectors & Polar Coordinates	Tests 9A, 9B, 9C, 9D (80 probs)	Level 1 → Level 4 (Grades 8–12+)	**pp. 22–23**
Section 10: Exponents, Logarithms & Growth Models	Tests 10A, 10B, 10C, 10D (80 probs)	Level 1 → Level 4 (Grades 8–12+)	**pp. 24–25**
Section 11: Combinatorics, Probability & Statistics	Tests 11A, 11B, 11C, 11D (80 probs)	Level 1 → Level 4 (Grades 7–12+)	**pp. 26–27**
Section 12: Base Conversions, Binary & Bitwise Logic	Tests 12A, 12B, 12C, 12D (80 probs)	Level 1 → Level 4 (Grades 7–12+)	**pp. 28–29**
Cumulative Review C3: All-Level Championship (Sections 1–12)	Tests C3A, C3B, C3C, C3D (80 probs)	Level 1 → Level 4 (Grades 7–12+)	**pp. 30–31**
Appendix A: Complete Self-Check Answer Keys	All 60 Tests (1,200 probs)	Quick-Grading Columns for Tests A–D	**pp. 32–36**
Appendix B: Mental Math & Stack Speed Tactics	SC-6 & SC-32	Five Inside-Out Verification Patterns	**p. 37**
Appendix C: Progress Graph & Mastery Certificate	Personal Best Log	Score Progression Chart & Certificate	**p. 38**

Preface: How to Use These Speed & Fluency Tests

Four progressive 20-problem practice tests per section—built for mental math, paper fluency, and stack verification.

---

1. FOUR PROGRESSIVE PRACTICE TESTS FOR EVERY SECTION (TEST A → B → C → D)

Instead of a single one-size-fits-all test, every section in this book provides four distinct 20-problem practice tests arranged in increasing difficulty across two facing pages:
- Test A (Level 1 — Foundation, 20 problems): Builds confident recall of core facts and single-step operations.
- Test B (Level 2 — Core Fluency, 20 problems): Introduces larger numbers, missing parts, and inverse relationships.
- Test C (Level 3 — Multi-Step Speed, 20 problems): Combines two or three operations requiring inside-out mathematical structure.
- Test D (Level 4 — Advanced Challenge, 20 problems): Reaches high-ceiling mental math and competition speed.

2. THREE WAYS TO USE THIS BOOK AT HOME OR IN THE CLASSROOM

Mode 1 — Progressive Mastery (Start at Level 1, Grow to Level 4):

Younger students or students learning a new topic start with Test A (Level 1) and Test B (Level 2), writing their answers directly on the lines. As confidence grows—or in the next grade—they tackle Test C (Level 3) and Test D (Level 4).

Example Stack Check: 2 4 ENTER 1 8 +

Mode 2 — Untimed Kitchen-Table Review & Independent Self-Check:

Use any 20-problem column untimed as a calm 5-minute Friday review alongside Singapore Math, Saxon, RightStart, AoPS, or school textbooks. Students write their answers on the lines, then verify on their StackCalc calculator or check Appendix A (pp. 32–36).

Example Stack Check: 1 4 ENTER 6 + 9 ENTER 4 - ÷

Mode 3 — Timed Speed Drills & Math Club / Competition Practice:

Set a timer for the suggested goal at the top of the column (3 to 6 minutes). Students race through the 20 problems mentally or with their stack calculator, recording their elapsed time and score out of 20 on the Master Tracker (Page 1).

Example Stack Check: 3 7 ENTER +

3. SIMPLE, CLEAR SCORING (20 PROBLEMS PER TEST)

- Standard Fluency Score (Out of 20): Count 1 point for each correct answer (Score: ___ / 20), or multiply by 5 for a 100-point percentage score (18 / 20 = 90%).
- Optional Contest Scoring: For students preparing for middle or high school math competitions, award +5 points per correct answer and deduct 2 points per incorrect answer (max score 100 points per 20-problem test).

Speed & Fluency Progress Tracker

Record your score (out of 20) and best completion time across all four difficulty levels (Tests A, B, C & D).

HOW TO LOG YOUR PROGRESS ACROSS LEVELS 1 TO 4

Each section has 4 progressive 20-problem tests: Test A (Level 1: Foundation), Test B (Level 2: Core Fluency), Test C (Level 3: Multi-Step Speed), and Test D (Level 4: Advanced Challenge). Write your score (___/20) and time as you complete each level!

Sec	Mathematical Topic	Test A (Lvl 1)	Test B (Lvl 2)	Test C (Lvl 3)	Test D (Lvl 4)	Pages
1	**Step Counting, Benchmarks & Sequences** Foundational to Advanced	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	2–3
2	**Number Bonds & Missing Addends** Foundational to Advanced	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	4–5
3	**Complements & Make-10 Bridging** Foundational to Advanced	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	6–7
4	**Doubles, Equal Groups & Squares** Foundational to Advanced	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	8–9
C1	**Number Sense (Sections 1–4)** Foundational to Advanced	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	10–11
5	**Multi-Step Operations & Grouping** Grades 3–8+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	12–13
6	**Fractions, Decimals & Reciprocals** Grades 4–9+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	14–15
7	**Primes, GCF/LCM & Nested Expressions** Grades 5–10+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	16–17
8	**Algebra, Quadratics & Coordinate Geometry** Grades 6–10+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	18–19
C2	**Intermediate Math (Sections 1–8)** Grades 5–10+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	20–21
9	**Trigonometry, Vectors & Polar Coordinates** Grades 8–12+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	22–23
10	**Exponents, Logarithms & Growth Models** Grades 8–12+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	24–25
11	**Combinatorics, Probability & Statistics** Grades 7–12+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	26–27
12	**Base Conversions, Binary & Bitwise Logic** Grades 7–12+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	28–29
C3	**All-Level Championship (Sections 1–12)** Grades 7–12+	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	30–31

Checking Your Answers

Turn to Appendix A (Pages 32–36) for the complete 60-test answer key, or verify any multi-step problem directly on your StackCalc SC-6 or SC-32 calculator.

Section 1: Step Counting, Benchmarks & Sequences

Unit steps, skip-counting, 10/100 benchmarks, and arithmetic series. • Foundation (Test 1A) & Core Fluency (Test 1B)

TEST 1A • LEVEL 1: FOUNDATION — Goal: 3 min

Adding & Subtracting 1, 2, and 3 Within 20

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$3 + 1 =$	
2.	$5 + 1 =$	
3.	$8 + 1 =$	
4.	$6 - 1 =$	
5.	$10 - 1 =$	
6.	$4 + 2 =$	
7.	$7 + 2 =$	
8.	$9 - 2 =$	
9.	$2 + 1 + 1 =$	
10.	$5 + 1 + 1 =$	
11.	$8 - 1 - 1 =$	
12.	$11 + 1 =$	
13.	$14 - 1 =$	
14.	$12 + 2 =$	
15.	$16 - 2 =$	
16.	$10 + 3 =$	
17.	$15 - 3 =$	
18.	$7 + 1 + 2 =$	
19.	$13 + 2 + 1 =$	
20.	$19 - 2 - 1 =$	

TEST 1B • LEVEL 2: CORE FLUENCY — Goal: 4 min

Skip-Counting Steps by 2s, 5s, and 10s

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$2 + 2 + 2 =$	
2.	$2 + 2 + 2 + 2 =$	
3.	$5 + 5 =$	
4.	$5 + 5 + 5 =$	
5.	$10 + 10 + 10 =$	
6.	$12 + 2 + 2 =$	
7.	$18 - 2 - 2 =$	
8.	$15 + 5 + 5 =$	
9.	$25 - 5 - 5 =$	
10.	$20 + 10 + 5 =$	
11.	$30 - 10 - 5 =$	
12.	$14 + 3 + 3 =$	
13.	$22 + 4 + 4 =$	
14.	$40 - 5 - 5 =$	
15.	$35 + 10 + 5 =$	
16.	$50 - 10 - 10 =$	
17.	$16 + 2 + 2 + 2 =$	
18.	$45 + 5 + 10 =$	
19.	$70 - 10 - 5 =$	
20.	$85 + 5 + 10 =$	

Section 1: Step Counting, Benchmarks & Sequences

Unit steps, skip-counting, 10/100 benchmarks, and arithmetic series. • Multi-Step Speed (Test 1C) & Advanced Challenge (Test 1D)

TEST 1C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Crossing 10s and 100s with Multi-Step Jumps

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$28 + 2 + 6 =$	_____
2.	$37 + 3 + 8 =$	_____
3.	$54 - 4 - 7 =$	_____
4.	$63 - 3 - 9 =$	_____
5.	$45 + 15 + 10 =$	_____
6.	$82 - 12 - 15 =$	_____
7.	$96 + 4 + 25 =$	_____
8.	$115 - 15 - 20 =$	_____
9.	$75 + 25 + 35 =$	_____
10.	$150 - 25 - 25 =$	_____
11.	$68 + 12 + 20 =$	_____
12.	$134 - 14 - 30 =$	_____
13.	$180 + 20 + 45 =$	_____
14.	$260 - 60 - 35 =$	_____
15.	$125 + 25 + 50 =$	_____
16.	$310 - 10 - 45 =$	_____
17.	$240 + 30 + 30 =$	_____
18.	$420 - 70 - 50 =$	_____
19.	$375 + 25 + 80 =$	_____
20.	$500 - 125 - 75 =$	_____

TEST 1D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Arithmetic Sequences & Fast Series Sums

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$1 + 2 + 3 + 4 + 5 =$	_____
2.	$2 + 4 + 6 + 8 + 10 =$	_____
3.	$1 + 3 + 5 + 7 + 9 =$	_____
4.	$3 + 6 + 9 + 12 =$	_____
5.	$4 + 8 + 12 + 16 =$	_____
6.	$5 + 10 + 15 + 20 + 25 =$	_____
7.	$1 + 2 + 3 + 4 + 5 + 6 + 7 =$	_____
8.	$10 + 12 + 14 + 16 =$	_____
9.	$15 + 20 + 25 + 30 =$	_____
10.	$11 + 13 + 15 + 17 =$	_____
11.	$20 + 25 + 30 + 35 + 40 =$	_____
12.	$1 + 3 + 5 + 7 + 9 + 11 =$	_____
13.	$6 + 12 + 18 + 24 =$	_____
14.	$100 - 12 - 18 - 20 =$	_____
15.	$25 + 50 + 75 + 100 =$	_____
16.	$14 + 16 + 18 + 20 + 22 =$	_____
17.	$7 + 14 + 21 + 28 =$	_____
18.	$50 + 60 + 70 + 80 + 90 =$	_____
19.	$1 + 2 + 3 + \ldots + 10 =$	_____
20.	$2 + 4 + 6 + \ldots + 20 =$	_____

Section 2: Number Bonds & Missing Addends

Part-part-whole composition, decomposition, and missing-part equations. • Foundation (Test 2A) & Core Fluency (Test 2B)

TEST 2A • LEVEL 1: FOUNDATION — Goal: 3 min

Combining & Decomposing Parts Within 12

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$4 + 3 =$	
2.	$5 + 2 =$	
3.	$3 + 5 =$	
4.	$6 + 2 =$	
5.	$4 + 5 =$	
6.	$7 + 2 =$	
7.	$6 + 3 =$	
8.	$7 - 3 =$	
9.	$8 - 5 =$	
10.	$9 - 4 =$	
11.	$9 - 6 =$	
12.	$5 + 6 =$	
13.	$7 + 4 =$	
14.	$8 + 3 =$	
15.	$7 + 5 =$	
16.	$8 + 4 =$	
17.	$9 + 3 =$	
18.	$11 - 5 =$	
19.	$12 - 7 =$	
20.	$12 - 8 =$	

TEST 2B • LEVEL 2: CORE FLUENCY — Goal: 4 min

Missing Addends & Subtraction Within 25

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$6 + 8 =$	
2.	$7 + 6 =$	
3.	$8 + 7 =$	
4.	$9 + 6 =$	
5.	$8 + 9 =$	
6.	$13 - 6 =$	
7.	$14 - 8 =$	
8.	$15 - 7 =$	
9.	$16 - 9 =$	
10.	$17 - 8 =$	
11.	$5 + [\ ?\ ] = 13$	
12.	$7 + [\ ?\ ] = 16$	
13.	$9 + [\ ?\ ] = 18$	
14.	$[\ ?\ ] + 6 = 15$	
15.	$[\ ?\ ] + 8 = 19$	
16.	$20 - 13 =$	
17.	$22 - 8 =$	
18.	$14 + [\ ?\ ] = 21$	
19.	$24 - 9 =$	
20.	$16 + [\ ?\ ] = 25$	

Section 2: Number Bonds & Missing Addends

Part-part-whole composition, decomposition, and missing-part equations. • Multi-Step Speed (Test 2C) & Advanced Challenge (Test 2D)

TEST 2C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Two-Digit Number Bonds & Missing Parts

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	24 + 15 =	_____
2.	32 + 27 =	_____
3.	46 + 18 =	_____
4.	38 + 25 =	_____
5.	54 - 21 =	_____
6.	60 - 24 =	_____
7.	72 - 35 =	_____
8.	81 - 46 =	_____
9.	28 + [?] = 50	_____
10.	34 + [?] = 60	_____
11.	[?] + 42 = 75	_____
12.	[?] + 37 = 80	_____
13.	90 - 56 =	_____
14.	85 - 29 =	_____
15.	47 + 36 =	_____
16.	58 + 37 =	_____
17.	63 + [?] = 100	_____
18.	100 - 48 =	_____
19.	110 - 65 =	_____
20.	74 + [?] = 120	_____

TEST 2D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Three-Part Bonds & Balanced Equations

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	125 + 65 =	_____
2.	240 - 85 =	_____
3.	135 + [?] = 200	_____
4.	168 + [?] = 250	_____
5.	320 - 145 =	_____
6.	48 + 35 + 52 =	_____
7.	64 + 29 + 36 =	_____
8.	150 - 45 - 35 =	_____
9.	200 - 68 - 32 =	_____
10.	85 + 45 = 100 + [?]	_____
11.	76 + 54 = 90 + [?]	_____
12.	140 - 55 = 50 + [?]	_____
13.	360 - 175 =	_____
14.	245 + 155 - 80 =	_____
15.	500 - 185 - 115 =	_____
16.	325 + [?] = 600	_____
17.	450 - [?] = 185	_____
18.	(165 + 85) - 70 =	_____
19.	(340 - 160) + 95 =	_____
20.	1000 - 425 - 175 =	_____

Section 3: Complements & Make-10 Bridging

Complements to 10, 100, and 1,000, bridging across tens, and compensation. • Foundation (Test 3A) & Core Fluency (Test 3B)

TEST 3A • LEVEL 1: FOUNDATION Goal: 3 min

Complements to 10 & Teen Place Value

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	10 - 7 =	
2.	10 - 4 =	
3.	10 - 8 =	
4.	10 - 3 =	
5.	10 - 9 =	
6.	10 - 6 =	
7.	10 - 2 =	
8.	10 - 5 =	
9.	10 + 4 =	
10.	10 + 7 =	
11.	10 + 9 =	
12.	8 + 2 + 3 =	
13.	9 + 1 + 5 =	
14.	7 + 3 + 4 =	
15.	6 + 4 + 2 =	
16.	8 + 5 =	
17.	9 + 6 =	
18.	7 + 5 =	
19.	8 + 6 =	
20.	9 + 7 =	

TEST 3B • LEVEL 2: CORE FLUENCY Goal: 4 min

Bridging Across 10, 20, and 50

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	9 + 8 =	
2.	8 + 7 =	
3.	7 + 6 =	
4.	9 + 5 =	
5.	20 - 7 =	
6.	20 - 4 =	
7.	20 - 12 =	
8.	18 + 5 =	
9.	19 + 7 =	
10.	17 + 6 =	
11.	28 + 6 =	
12.	29 + 8 =	
13.	37 + 5 =	
14.	30 - 6 =	
15.	40 - 8 =	
16.	50 - 14 =	
17.	50 - 27 =	
18.	48 + 7 =	
19.	59 + 8 =	
20.	70 - 23 =	

Section 3: Complements & Make-10 Bridging

Complements to 10, 100, and 1,000, bridging across tens, and compensation. • Multi-Step Speed (Test 3C) & Advanced Challenge (Test 3D)

TEST 3C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Complements to 100 & Fast Compensation (+9, +19, +29)

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	**100 - 30** =	_____
2.	**100 - 45** =	_____
3.	**100 - 62** =	_____
4.	**100 - 28** =	_____
5.	**100 - 74** =	_____
6.	**100 - 19** =	_____
7.	**100 - 83** =	_____
8.	**100 - 57** =	_____
9.	**36 + 9** =	_____
10.	**58 + 9** =	_____
11.	**47 + 19** =	_____
12.	**65 + 19** =	_____
13.	**38 + 29** =	_____
14.	**54 + 39** =	_____
15.	**73 - 9** =	_____
16.	**82 - 19** =	_____
17.	**95 - 29** =	_____
18.	**67 + 33 + 25** =	_____
19.	**48 + 52 + 39** =	_____
20.	**200 - 64** =	_____

TEST 3D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Complements to 1,000, Decimals & Near-Hundred Sums

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	**1000 - 400** =	_____
2.	**1000 - 650** =	_____
3.	**1000 - 275** =	_____
4.	**1000 - 384** =	_____
5.	**1000 - 742** =	_____
6.	**1000 - 519** =	_____
7.	**1000 - 863** =	_____
8.	**1000 - 97** =	_____
9.	**10 - 3.4** =	_____
10.	**10 - 6.75** =	_____
11.	**100 - 42.5** =	_____
12.	**99 + 47** =	_____
13.	**98 + 65** =	_____
14.	**199 + 84** =	_____
15.	**299 + 156** =	_____
16.	**498 + 237** =	_____
17.	**99 + 99 + 99** =	_____
18.	**98 + 97 + 96** =	_____
19.	**198 + 297 + 105** =	_____
20.	**5000 - 1645** =	_____

Section 4: Doubles, Equal Groups & Squares

Doubles, near-doubles, equal groups, halving, and mental squaring. • Foundation (Test 4A) & Core Fluency (Test 4B)

TEST 4A • LEVEL 1: FOUNDATION — Goal: 3 min

Doubles and Near-Doubles Within 25

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$3 + 3 =$	
2.	$4 + 4 =$	
3.	$5 + 5 =$	
4.	$6 + 6 =$	
5.	$7 + 7 =$	
6.	$8 + 8 =$	
7.	$9 + 9 =$	
8.	$10 + 10 =$	
9.	$4 + 5 =$	
10.	$5 + 6 =$	
11.	$6 + 7 =$	
12.	$7 + 8 =$	
13.	$8 + 9 =$	
14.	$9 + 10 =$	
15.	$11 + 11 =$	
16.	$12 + 12 =$	
17.	$10 + 11 =$	
18.	$11 + 12 =$	
19.	$2 \times 6 =$	
20.	$2 \times 9 =$	

TEST 4B • LEVEL 2: CORE FLUENCY — Goal: 4 min

Two-Digit Doubles, Halves & Equal Groups

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$13 + 13 =$	
2.	$14 + 14 =$	
3.	$15 + 15 =$	
4.	$16 + 16 =$	
5.	$18 + 18 =$	
6.	$25 + 25 =$	
7.	$35 + 35 =$	
8.	$45 + 45 =$	
9.	$15 + 16 =$	
10.	$24 + 25 =$	
11.	$3 \times 4 =$	
12.	$4 \times 5 =$	
13.	$3 \times 6 =$	
14.	$5 \times 6 =$	
15.	$4 \times 8 =$	
16.	$6 \times 7 =$	
17.	$24 \div 2 =$	
18.	$36 \div 2 =$	
19.	$50 \div 2 =$	
20.	$48 \div 2 =$	

Section 4: Doubles, Equal Groups & Squares

Doubles, near-doubles, equal groups, halving, and mental squaring. • Multi-Step Speed (Test 4C) & Advanced Challenge (Test 4D)

TEST 4C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Doubling Chains (×2, ×4, ×8) & Two-Digit Multiplication

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$2 \times 27 =$	
2.	$2 \times 38 =$	
3.	$2 \times 46 =$	
4.	$2 \times 65 =$	
5.	$2 \times 78 =$	
6.	$4 \times 15 =$	
7.	$4 \times 18 =$	
8.	$4 \times 23 =$	
9.	$4 \times 25 =$	
10.	$4 \times 35 =$	
11.	$8 \times 12 =$	
12.	$8 \times 15 =$	
13.	$8 \times 25 =$	
14.	$74 \div 2 =$	
15.	$96 \div 2 =$	
16.	$130 \div 2 =$	
17.	$84 \div 4 =$	
18.	$120 \div 4 =$	
19.	$2 \times 34 + 10 =$	
20.	$4 \times 16 - 14 =$	

TEST 4D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Mental Squares, Near-Squares (a-1)(a+1) & Powers

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$11^2 =$	
2.	$12^2 =$	
3.	$13^2 =$	
4.	$14^2 =$	
5.	$15^2 =$	
6.	$16^2 =$	
7.	$20^2 =$	
8.	$25^2 =$	
9.	$30^2 =$	
10.	$35^2 =$	
11.	$9 \times 11 =$	
12.	$11 \times 13 =$	
13.	$14 \times 16 =$	
14.	$19 \times 21 =$	
15.	$24 \times 26 =$	
16.	$29 \times 31 =$	
17.	$2 \times 15^2 =$	
18.	$10^2 + 12^2 =$	
19.	$15^2 - 5^2 =$	
20.	$25^2 - 15^2 =$	

Cumulative Review C1: Number Sense (Sections 1–4)

Integrated review across step counting, number bonds, complements, and doubles. • Foundation (Test C1A) & Core Fluency (Test C1B)

TEST C1A • LEVEL 1: FOUNDATION Goal: 4 min

Core Addition, Subtraction, Complements to 10 & Doubles

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$6 + 1 + 1 =$	
2.	$12 - 2 =$	
3.	$5 + 4 =$	
4.	$11 - 6 =$	
5.	$10 - 8 =$	
6.	$10 - 3 =$	
7.	$10 + 6 =$	
8.	$7 + 7 =$	
9.	$8 + 8 =$	
10.	$6 + 7 =$	
11.	$8 + 5 =$	
12.	$9 + 6 =$	
13.	$14 - 7 =$	
14.	$16 - 8 =$	
15.	$5 + 5 + 5 =$	
16.	$9 + 9 =$	
17.	$7 + 8 =$	
18.	$15 - 6 =$	
19.	$10 + 10 =$	
20.	$20 - 5 =$	

TEST C1B • LEVEL 2: CORE FLUENCY Goal: 4 min

Two-Digit Fluency, Missing Addends & Equal Groups

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$18 + 2 + 7 =$	
2.	$35 - 5 - 10 =$	
3.	$8 + [\ ?\ ] = 17$	
4.	$14 + [\ ?\ ] = 22$	
5.	$30 - 14 =$	
6.	$50 - 18 =$	
7.	$16 + 16 =$	
8.	$24 + 24 =$	
9.	$17 + 18 =$	
10.	$3 \times 7 =$	
11.	$4 \times 6 =$	
12.	$5 \times 8 =$	
13.	$32 \div 2 =$	
14.	$46 \div 2 =$	
15.	$29 + 7 =$	
16.	$48 + 9 =$	
17.	$60 - 25 =$	
18.	$25 + 25 + 15 =$	
19.	$4 \times 9 =$	
20.	$100 - 45 =$	

Cumulative Review C1: Number Sense (Sections 1–4)

Integrated review across step counting, number bonds, complements, and doubles. • Multi-Step Speed (Test C1C) & Advanced Challenge (Test C1D)

TEST C1C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Complements to 100, Doubling Chains & Multi-Step Bonds

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$100 - 37 =$	_____
2.	$100 - 68 =$	_____
3.	$46 + 38 =$	_____
4.	$83 - 47 =$	_____
5.	$57 + 19 =$	_____
6.	$84 - 29 =$	_____
7.	$2 \times 48 =$	_____
8.	$4 \times 19 =$	_____
9.	$8 \times 15 =$	_____
10.	$110 \div 2 =$	_____
11.	$92 \div 4 =$	_____
12.	$64 + [\ ?\ ] = 120$	_____
13.	$150 - 85 =$	_____
14.	$35 + 45 + 20 =$	_____
15.	$12^2 =$	_____
16.	$15^2 =$	_____
17.	$2 \times 36 + 18 =$	_____
18.	$100 - 24 - 36 =$	_____
19.	$250 - 125 + 40 =$	_____
20.	$4 \times 25 - 33 =$	_____

TEST C1D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Series Sums, Near-Squares & Mental Compensation

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$1 + 2 + 3 + 4 + 5 + 6 =$	_____
2.	$2 + 4 + 6 + 8 + 10 + 12 =$	_____
3.	$1000 - 465 =$	_____
4.	$1000 - 782 =$	_____
5.	$199 + 247 =$	_____
6.	$398 + 165 =$	_____
7.	$14^2 =$	_____
8.	$25^2 =$	_____
9.	$19 \times 21 =$	_____
10.	$29 \times 31 =$	_____
11.	$10 - 4.35 =$	_____
12.	$100 - 63.8 =$	_____
13.	$99 + 98 + 97 + 96 =$	_____
14.	$18^2 - 8^2 =$	_____
15.	$45 + 55 + 65 + 75 =$	_____
16.	$8 \times 35 =$	_____
17.	$420 \div 4 =$	_____
18.	$350 + [\ ?\ ] = 825$	_____
19.	$2 \times 20^2 - 150 =$	_____
20.	$1 + 3 + 5 + 7 + 9 + 11 + 13 =$	_____

Section 5: Multi-Step Operations & Grouping

Multiplication, division, inside-out parentheses, and two-branch expressions. • Foundation (Test 5A) & Core Fluency (Test 5B)

TEST 5A • LEVEL 1: FOUNDATION Goal: 4 min

Core Multiplication, Division & Two-Step Expressions

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$6 \times 4 =$	
2.	$7 \times 5 =$	
3.	$8 \times 6 =$	
4.	$9 \times 7 =$	
5.	$24 \div 3 =$	
6.	$36 \div 4 =$	
7.	$45 \div 5 =$	
8.	$56 \div 8 =$	
9.	$(10 - 4) \times 3 =$	
10.	$(12 + 8) \div 4 =$	
11.	$(15 - 6) \times 2 =$	
12.	$(18 + 6) \div 3 =$	
13.	$5 \times 4 + 7 =$	
14.	$6 \times 5 - 8 =$	
15.	$20 - 3 \times 4 =$	
16.	$30 - 4 \times 5 =$	
17.	$14 + 18 \div 3 =$	
18.	$25 - 20 \div 4 =$	
19.	$(9 + 6) \times 2 =$	
20.	$40 \div (3 + 5) =$	

TEST 5B • LEVEL 2: CORE FLUENCY Goal: 4 min

Order of Operations & Inner Parentheses

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$(18 - 6) \times 3 =$	
2.	$(24 + 16) \div 8 =$	
3.	$45 \div (12 - 3) =$	
4.	$60 \div (7 + 5) =$	
5.	$12 + 6 \times 4 =$	
6.	$50 - 7 \times 6 =$	
7.	$8 \times 7 - 16 =$	
8.	$9 \times 8 + 18 =$	
9.	$(32 - 12) \times 4 =$	
10.	$(48 + 12) \div 6 =$	
11.	$72 \div (4 + 5) =$	
12.	$100 - 15 \times 4 =$	
13.	$24 + 36 \div 6 =$	
14.	$64 \div 8 + 17 =$	
15.	$(14 + 7) \div 3 \times 5 =$	
16.	$(25 - 10) \times 4 \div 6 =$	
17.	$90 \div (15 - 6) \times 4 =$	
18.	$120 - (8 \times 9) =$	
19.	$(16 + 24) \div (12 - 4) =$	
20.	$(15 - 3) \times (2 + 4) =$	

Section 5: Multi-Step Operations & Grouping

Multiplication, division, inside-out parentheses, and two-branch expressions. • Multi-Step Speed (Test 5C) & Advanced Challenge (Test 5D)

TEST 5C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Two-Branch Expressions (a ± b) × (c ± d) & Quotients

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	(15 × 4) + (18 ÷ 2) =	
2.	(12 × 5) - (8 × 4) =	
3.	(50 - 14) ÷ (3 + 6) =	
4.	(42 + 18) ÷ (11 - 5) =	
5.	(8 + 4) × (9 - 3) =	
6.	(14 - 5) × (6 + 2) =	
7.	(20 × 3) + (15 × 2) =	
8.	(16 × 4) - (12 × 3) =	
9.	((20 - 8) × 4) ÷ 6 =	
10.	((18 + 6) ÷ 3) × 7 =	
11.	(72 ÷ (4 + 5)) × 3 =	
12.	100 - (16 × 5 - 12) =	
13.	(84 ÷ 7) + (45 ÷ 9) =	
14.	(65 - 25) × (18 ÷ 6) =	
15.	((8 + 4) × (9 - 3)) ÷ 12 =	
16.	(96 ÷ 8) × (20 - 15) =	
17.	150 - (12 × 6 + 18) =	
18.	(35 + 25) ÷ (4 × 3) =	
19.	(18 × 5) ÷ (14 - 5) =	
20.	(24 - 9) × (16 - 8) =	

TEST 5D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Multi-Digit Chained Operations & Deep Nesting

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	(25 × 8) - (15 × 6) =	
2.	(144 ÷ 12) × (36 ÷ 4) =	
3.	(125 + 75) ÷ (32 - 7) =	
4.	((45 - 15) × 6) ÷ (12 + 6) =	
5.	250 - (18 × 7 + 24) =	
6.	(36 × 5) ÷ (27 - 9) =	
7.	((64 ÷ 4) + 9) × 4 =	
8.	(180 ÷ (12 + 3)) × 8 =	
9.	(22 + 18) × (35 - 20) =	
10.	(160 - 40) ÷ (6 × 4) =	
11.	((32 - 14) × 5) ÷ 6 =	
12.	80 - (15 × 4) + 12 =	
13.	(240 ÷ 15) + (120 ÷ 8) =	
14.	(17 × 6) - (13 × 6) =	
15.	(28 × 9) - (18 × 9) =	
16.	(45 × 12) ÷ 18 =	
17.	((100 - 28) ÷ 9) × 15 =	
18.	(360 ÷ (18 - 6)) - 14 =	
19.	((15 + 5) × (8 - 3)) ÷ 10 =	
20.	(500 - 140) ÷ (9 × 4) =	

Section 6: Fractions, Decimals & Reciprocals

Fraction equivalence, unlike denominators, mixed numbers, decimals, and reciprocals. • Foundation (Test 6A) & Core Fluency (Test 6B)

TEST 6A • LEVEL 1: FOUNDATION Goal: 4 min

Like Fractions, Halves/Fourths & Exact Decimals

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **1/4 + 2/4 =** _____
2. **3/5 - 1/5 =** _____
3. **3/8 + 2/8 =** _____
4. **7/10 - 4/10 =** _____
5. **1/2 + 1/4 =** _____
6. **3/4 - 1/2 =** _____
7. **1/2 + 1/2 =** _____
8. **1 - 1/4 =** _____
9. **1 - 3/8 =** _____
10. **1/2 as a decimal =** _____
11. **1/4 as a decimal =** _____
12. **3/4 as a decimal =** _____
13. **1/5 as a decimal =** _____
14. **3/5 as a decimal =** _____
15. **1/10 + 3/10 (simplified) =** _____
16. **2/6 + 1/6 (simplified) =** _____
17. **1/2 of 16 =** _____
18. **1/4 of 20 =** _____
19. **3/4 of 12 =** _____
20. **2/3 of 15 =** _____

TEST 6B • LEVEL 2: CORE FLUENCY Goal: 5 min

Unlike Denominators, Multiplication & Reciprocals

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **1/3 + 1/6 =** _____
2. **2/3 - 1/6 =** _____
3. **3/8 + 1/4 =** _____
4. **5/6 - 1/3 =** _____
5. **1/2 + 1/3 =** _____
6. **1/2 - 1/3 =** _____
7. **1/3 + 1/4 =** _____
8. **3/4 - 1/3 =** _____
9. **1/2 × 3/4 =** _____
10. **2/3 × 3/4 =** _____
11. **3/5 × 5/6 =** _____
12. **4/5 × 15 =** _____
13. **Reciprocal of 3/4 =** _____
14. **Reciprocal of 4/5 (decimal) =** _____
15. **Reciprocal of 2/5 (decimal) =** _____
16. **1/2 ÷ 1/4 =** _____
17. **3/4 ÷ 1/4 =** _____
18. **3/4 ÷ 1/2 (decimal) =** _____
19. **2/3 ÷ 2/3 =** _____
20. **6 ÷ 1/2 =** _____

Section 6: Fractions, Decimals & Reciprocals

Fraction equivalence, unlike denominators, mixed numbers, decimals, and reciprocals. • Multi-Step Speed (Test 6C) & Advanced Challenge (Test 6D)

TEST 6C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Mixed Numbers, Decimal Equivalents & Fraction Chains

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **2 1/2 + 1 1/4 (decimal) =** _____
2. **3 3/4 - 1 1/2 (decimal) =** _____
3. **5 - 2 3/4 (decimal) =** _____
4. **3 1/2 - 1 3/4 (decimal) =** _____
5. **1 1/2 × 4 =** _____
6. **2 1/2 × 6 =** _____
7. **1 1/4 × 8 =** _____
8. **3 1/3 × 3 =** _____
9. **(1/3 + 1/6) × 12 =** _____
10. **(1/2 + 1/4) × 8 =** _____
11. **(3/4 - 1/8) × 16 =** _____
12. **3/4 + (1/2 × 1/4) (decimal) =** _____
13. **1/8 as a decimal =** _____
14. **3/8 as a decimal =** _____
15. **5/8 as a decimal =** _____
16. **7/8 as a decimal =** _____
17. **4 1/2 ÷ 1 1/2 =** _____
18. **2 1/4 ÷ 3/4 =** _____
19. **0.25 + 0.5 + 0.75 =** _____
20. **1.25 × 8 =** _____

TEST 6D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Harmonic Means, Complex Fractions & Multi-Step Rationals

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **1 ÷ (1/2 + 1/2) =** _____
2. **1 ÷ (1/3 + 1/6) =** _____
3. **1 ÷ (1/4 + 1/4) =** _____
4. **1 ÷ (1/4 + 1/6) (decimal) =** _____
5. **1 ÷ (1/3 + 1/3 + 1/3) =** _____
6. **(1/2 + 1/3) × 12 =** _____
7. **(1/3 + 1/4) × 24 =** _____
8. **(3/4 - 2/3) × 12 =** _____
9. **(2/3 + 3/4) × 12 =** _____
10. **(1/2 + 1/3 + 1/6) × 18 =** _____
11. **2/3 × 3/4 × 4/5 =** _____
12. **1/2 × 2/3 × 3/4 × 4/5 =** _____
13. **(3/4) ÷ (3/8) =** _____
14. **(5/6) ÷ (5/12) =** _____
15. **(2 1/2 + 1 1/2) ÷ (1/4) =** _____
16. **1 + 1/(1 + 1/2) =** _____
17. **1 + 1/(1 + 1/3) =** _____
18. **(3 1/3 ÷ 1 1/3) (decimal) =** _____
19. **24 × (5/6 - 3/8) =** _____
20. **36 × (7/9 - 1/4) =** _____

Section 7: Primes, GCF/LCM & Nested Expressions

Prime factorization, GCF, LCM, powers, and multi-branch expression trees. • Foundation (Test 7A) & Core Fluency (Test 7B)

TEST 7A • LEVEL 1: FOUNDATION Goal: 4 min

Prime Powers, Squares, Cubes & Factor Pairs

Date: _____ Time: _____ Score: _____ / 20

1.	$2 \times 2 \times 3 =$	_____
2.	$2 \times 3 \times 3 =$	_____
3.	$2 \times 2 \times 5 =$	_____
4.	$2 \times 2 \times 2 \times 3 =$	_____
5.	$2 \times 3 \times 5 =$	_____
6.	$2 \times 2 \times 3 \times 3 =$	_____
7.	$2^3 =$	_____
8.	$2^4 =$	_____
9.	$3^3 =$	_____
10.	$4^3 =$	_____
11.	$5^3 =$	_____
12.	Smallest prime number =	_____
13.	7th prime (2,3,5,7,11,13,?) =	_____
14.	GCF of 8 and 12 =	_____
15.	GCF of 15 and 25 =	_____
16.	GCF of 18 and 24 =	_____
17.	LCM of 3 and 4 =	_____
18.	LCM of 4 and 6 =	_____
19.	LCM of 6 and 8 =	_____
20.	LCM of 5 and 10 =	_____

TEST 7B • LEVEL 2: CORE FLUENCY Goal: 5 min

GCF, LCM & Order of Operations Without Parentheses

Date: _____ Time: _____ Score: _____ / 20

1.	GCF of 24 and 36 =	_____
2.	GCF of 30 and 45 =	_____
3.	GCF of 32 and 48 =	_____
4.	GCF of 42 and 56 =	_____
5.	LCM of 8 and 12 =	_____
6.	LCM of 9 and 12 =	_____
7.	LCM of 12 and 15 =	_____
8.	LCM of 10 and 14 =	_____
9.	$10 + 5 \times 2 =$	_____
10.	$25 - 5 \times 3 =$	_____
11.	$4 + 6 \div 2 =$	_____
12.	$12 - 2 \times 4 =$	_____
13.	$15 - 3 \times 3 =$	_____
14.	$30 - 4 \times 5 =$	_____
15.	$2^3 \times 3^2 =$	_____
16.	$2^2 \times 3 \times 5 =$	_____
17.	$2^3 \times 5^2 =$	_____
18.	$3^2 + 4^2 =$	_____
19.	$5^2 - 3^2 =$	_____
20.	$10^2 - 8^2 =$	_____

Section 7: Primes, GCF/LCM & Nested Expressions

Prime factorization, GCF, LCM, powers, and multi-branch expression trees. • Multi-Step Speed (Test 7C) & Advanced Challenge (Test 7D)

TEST 7C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Nested Two-Branch Expression Trees

Date: _____ **Time:** _____ **Score:** _____ / 20

1. $(8 + 2) \times (9 - 4) =$ _____
2. $((14 - 6) \times 3) + (24 \div 4) =$ _____
3. $(30 - (4 \times 5)) \times (12 \div (2 + 4)) =$ _____
4. $((5 + 3) \times (8 - 2)) \div (6 - 2) =$ _____
5. $((18 \div 3) + 4) \times ((15 - 7) \div 2) =$ _____
6. $(10 + 5 \times 2) \div (12 - 2 \times 4) =$ _____
7. $(25 - 5 \times 3) \times (4 + 6 \div 2) =$ _____
8. $((7 + 3) \times 4 - 8) \div 4 =$ _____
9. $(48 \div (2 + 6)) \times (15 - 3 \times 3) =$ _____
10. $(16 - (4 + 2)) \times (18 \div (3 \times 2)) =$ _____
11. $(3^2 + 1) \times (4^2 - 11) =$ _____
12. $(5^2 - 5) \div (2^3 - 4) =$ _____
13. $((20 - 5) \div 3) \times (12 - 4) =$ _____
14. $(36 \div (3 \times 3)) + (5 \times 4) =$ _____
15. $(50 - 2 \times 15) \div (3 + 2) =$ _____
16. $((9 + 3) \times 5) \div (10 - 4) =$ _____
17. $(2^4 + 4) \times (3^2 - 6) =$ _____
18. $(100 - 6^2) \div (2^3) =$ _____
19. $(7^2 - 9) \div (2^2 + 4) =$ _____
20. $((12 + 8) \div 4)^3 =$ _____

TEST 7D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Number Theory: GCF × LCM, Divisor Counts & Deep Trees

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **GCF(12, 18) × LCM(12, 18) =** _____
2. **GCF(15, 20) × LCM(15, 20) =** _____
3. **GCF of 72, 108, and 144 =** _____
4. **LCM of 6, 8, and 12 =** _____
5. **LCM of 4, 9, and 12 =** _____
6. **LCM of 5, 6, and 8 =** _____
7. **Number of positive factors of 12 =** _____
8. **Number of positive factors of 24 =** _____
9. **Number of positive factors of 36 =** _____
10. **Sum of prime factors of 30 (2+3+5) =** _____
11. **Sum of all positive factors of 12 =** _____
12. $(6^2 - 4^2) \div (6 - 4) =$ _____
13. $(15^2 - 5^2) \div (15 + 5) =$ _____
14. $(25^2 - 15^2) \div (25 - 15) =$ _____
15. $((35 - 7) \times 4) \div (14 + 2) =$ _____
16. $(2^5 - 2^3) \div (3^2 - 3) =$ _____
17. $(3^4 - 1) \div (4^2 + 0) =$ _____
18. $(10^3 - 200) \div (4^2 \times 5) =$ _____
19. $((8^2 - 4) \div 15) \times (5^2 - 10) =$ _____
20. $(12^2 + 5^2) \div 13 =$ _____

Section 8: Algebra, Quadratics & Coordinate Geometry

Linear equations, slope, Pythagorean distance, and quadratic roots. • Foundation (Test 8A) & Core Fluency (Test 8B)

TEST 8A • LEVEL 1: FOUNDATION Goal: 4 min

Evaluating Expressions & Solving Linear Equations

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	**$3x + 5$ when $x = 4$:**	_____
2.	**$4x - 7$ when $x = 5$:**	_____
3.	**$2x^2$ when $x = 3$:**	_____
4.	**$x^2 + 4x$ when $x = 2$:**	_____
5.	**$5a - 2b$ when $a = 4$, $b = 3$:**	_____
6.	**Solve: $x + 9 = 21$**	_____
7.	**Solve: $x - 8 = 15$**	_____
8.	**Solve: $4x = 36$**	_____
9.	**Solve: $x \div 5 = 7$**	_____
10.	**Solve: $2x + 3 = 15$**	_____
11.	**Solve: $3x - 4 = 14$**	_____
12.	**Solve: $5x + 10 = 45$**	_____
13.	**Solve: $4x - 9 = 11$**	_____
14.	**Solve: $6x + 2 = 32$**	_____
15.	**Solve: $(x \div 3) + 2 = 6$**	_____
16.	**Evaluate $-8 + 15 =$**	_____
17.	**Evaluate $6 - 14 =$**	_____
18.	**Evaluate $(-4) \times (-6) =$**	_____
19.	**Evaluate $(-36) \div 9 =$**	_____
20.	**Evaluate $(-5)^2 =$**	_____

TEST 8B • LEVEL 2: CORE FLUENCY Goal: 5 min

Slope, Midpoints & Pythagorean Right Triangles

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	**Slope from (0, 0) to (4, 12) =**	_____
2.	**Slope from (1, 2) to (5, 10) =**	_____
3.	**Slope from (2, 3) to (6, 15) =**	_____
4.	**Slope from (-1, -2) to (3, 10) =**	_____
5.	**Slope from (0, 10) to (5, 0) =**	_____
6.	**Slope from (2, 8) to (6, 6) =**	_____
7.	**Midpoint x-value of (2, 5) & (8, 9) =**	_____
8.	**Midpoint y-value of (2, 5) & (8, 9) =**	_____
9.	**Hypotenuse for legs 3 and 4 =**	_____
10.	**Hypotenuse for legs 6 and 8 =**	_____
11.	**Hypotenuse for legs 5 and 12 =**	_____
12.	**Hypotenuse for legs 8 and 15 =**	_____
13.	**Hypotenuse for legs 9 and 12 =**	_____
14.	**Hypotenuse for legs 12 and 16 =**	_____
15.	**Missing leg: hyp = 10, leg = 6 →**	_____
16.	**Missing leg: hyp = 13, leg = 5 →**	_____
17.	**Distance from (0, 0) to (3, 4) =**	_____
18.	**Distance from (1, 1) to (4, 5) =**	_____
19.	**Distance from (3, 4) to (9, 12) =**	_____
20.	**Distance from (-3, 5) to (5, -10) =**	_____

Section 8: Algebra, Quadratics & Coordinate Geometry

Linear equations, slope, Pythagorean distance, and quadratic roots. • Multi-Step Speed (Test 8C) & Advanced Challenge (Test 8D)

TEST 8C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Polynomial Evaluation & Quadratic Factoring

Date: _____ Time: _____ Score: _____ / 20

1. Evaluate $x^2 + 5x + 6$ at $x = 2$: _____
2. Evaluate $2x^2 + 4x - 5$ at $x = 3$: _____
3. Evaluate $x^2 - 6x + 9$ at $x = 5$: _____
4. Evaluate $3x^2 - 2x + 1$ at $x = -2$: _____
5. Greater root of $x^2 + 5x + 6 = 0$: _____
6. Lesser root of $x^2 + 5x + 6 = 0$: _____
7. Greater root of $x^2 + 7x + 12 = 0$: _____
8. Greater root of $x^2 + 8x + 15 = 0$: _____
9. Positive root of $x^2 - x - 6 = 0$: _____
10. Lesser root of $x^2 - 5x + 6 = 0$: _____
11. Greater root of $x^2 - 5x + 6 = 0$: _____
12. Positive root of $x^2 - 7x + 10 = 0$ (max) : _____
13. Positive root of $x^2 - 9 = 0$: _____
14. Positive root of $x^2 - 16 = 0$: _____
15. Positive root of $x^2 - 25 = 0$: _____
16. Repeated root of $x^2 + 10x + 25 = 0$: _____
17. Repeated root of $x^2 - 8x + 16 = 0$: _____
18. Greater root of $2x^2 + 5x + 2 = 0$: _____
19. Positive root of $x^2 + 2x - 15 = 0$: _____
20. Positive root of $x^2 - 3x - 10 = 0$: _____

TEST 8D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Vieta's Root Sums/Products, Linear Systems & 3D Distance

Date: _____ Time: _____ Score: _____ / 20

1. Sum of roots of $x^2 - 9x + 20 = 0$: _____
2. Product of roots of $x^2 - 9x + 20 = 0$: _____
3. Sum of roots of $x^2 + 11x + 24 = 0$: _____
4. Product of roots of $x^2 + 11x + 24 = 0$: _____
5. Sum of roots of $2x^2 - 10x + 7 = 0$: _____
6. Product of roots of $2x^2 - 10x + 7 = 0$: _____
7. Discriminant $b^2 - 4ac$ of $x^2 + 6x + 5$: _____
8. Discriminant $b^2 - 4ac$ of $x^2 - 4x + 4$: _____
9. Vertex x-coord of $y = x^2 - 8x + 19$: _____
10. Minimum value of $y = x^2 - 8x + 19$: _____
11. Solve for x: $x + y = 10$, $x - y = 4$: _____
12. Solve for y: $x + y = 10$, $x - y = 4$: _____
13. Solve for x: $2x + y = 13$, $x + y = 8$: _____
14. Solve for x: $3x - 4 = x + 10$: _____
15. Solve for x: $5x + 2 = 2x + 17$: _____
16. Hypotenuse for legs 7 and 24 = _____
17. Hypotenuse for legs 9 and 40 = _____
18. 3D diagonal $\sqrt{(3^2 + 4^2 + 12^2)}$ = _____
19. 3D diagonal $\sqrt{(2^2 + 3^2 + 6^2)}$ = _____
20. 3D diagonal $\sqrt{(1^2 + 4^2 + 8^2)}$ = _____

Cumulative Review C2: Intermediate Math (Sections 1–8)

Comprehensive review across fractions, expressions, number theory, and algebra. • Foundation (Test C2A) & Core Fluency (Test C2B)

TEST C2A • LEVEL 1: FOUNDATION Goal: 4 min

Multi-Step Arithmetic, Fractions & Basic Algebra

Date: _____ Time: _____ Score: _____ / 20

1.	18 + 18 - 9 =	_____
2.	100 - 36 =	_____
3.	(18 - 6) × 3 =	_____
4.	(24 + 16) ÷ 8 =	_____
5.	1/2 + 1/4 (decimal) =	_____
6.	3/4 - 1/2 (decimal) =	_____
7.	2/3 of 18 =	_____
8.	3/4 of 24 =	_____
9.	$2^3 \times 3$ =	_____
10.	GCF of 12 and 18 =	_____
11.	LCM of 4 and 6 =	_____
12.	10 + 5 × 2 =	_____
13.	30 - 4 × 5 =	_____
14.	Solve: x + 12 = 30	_____
15.	Solve: 3x = 42	_____
16.	Solve: 2x + 5 = 19	_____
17.	Evaluate 3x - 4 at x = 6 :	_____
18.	Hypotenuse for legs 3 and 4 =	_____
19.	Hypotenuse for legs 6 and 8 =	_____
20.	Slope from (0, 0) to (3, 12) =	_____

TEST C2B • LEVEL 2: CORE FLUENCY Goal: 5 min

Rational Operations, Expression Trees & Coordinate Slope

Date: _____ Time: _____ Score: _____ / 20

1.	3 1/2 - 1 3/4 (decimal) =	_____
2.	2 1/2 + 1 1/4 (decimal) =	_____
3.	Reciprocal of 4/5 (decimal) =	_____
4.	3/4 ÷ 1/2 (decimal) =	_____
5.	((32 - 14) × 5) ÷ 6 =	_____
6.	((15 + 5) × (8 - 3)) ÷ 10 =	_____
7.	80 - (15 × 4) + 12 =	_____
8.	(8 + 2) × (9 - 4) =	_____
9.	GCF of 24 and 36 =	_____
10.	LCM of 8 and 12 =	_____
11.	Slope from (-1, -2) to (3, 10) =	_____
12.	Distance from (0, 0) to (12, 16) =	_____
13.	Distance from (3, 4) to (9, 12) =	_____
14.	Positive root of $x^2 - 9 = 0$:	_____
15.	Positive root of $x^2 - 16 = 0$:	_____
16.	Greater root of $x^2 + 5x + 6 = 0$:	_____
17.	Lesser root of $x^2 - 5x + 6 = 0$:	_____
18.	Evaluate $x^2 + 3x$ at x = 4 :	_____
19.	15^2 =	_____
20.	25^2 =	_____

Cumulative Review C2: Intermediate Math (Sections 1–8)

Comprehensive review across fractions, expressions, number theory, and algebra. • Multi-Step Speed (Test C2C) & Advanced Challenge (Test C2D)

TEST C2C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Complex Fractions, Quadratics & Multi-Branch Evaluation

Date: _____ Time: _____ Score: _____ / 20

1. $3/4 + (1/2 \times 1/4)$ (decimal) = _____
2. $1 \div (1/4 + 1/6)$ (decimal) = _____
3. $(1/3 + 1/4) \times 24$ = _____
4. $(2/3 + 3/4) \times 12$ = _____
5. $((5 + 3) \times (8 - 2)) \div (6 - 2)$ = _____
6. $((18 \div 3) + 4) \times ((15 - 7) \div 2)$ = _____
7. $(25 - 5 \times 3) \times (4 + 6 \div 2)$ = _____
8. Evaluate $2x^2 + 4x - 5$ at $x = 3$: _____
9. Positive root of $x^2 - x - 6 = 0$: _____
10. Greater root of $2x^2 + 5x + 2 = 0$: _____
11. Repeated root of $x^2 + 10x + 25 = 0$: _____
12. Sum of roots of $x^2 - 7x + 12 = 0$: _____
13. Product of roots of $x^2 - 7x + 12 = 0$: _____
14. Solve: $4x - 5 = 2x + 9$ _____
15. Missing leg: hyp = 13, leg = 5 → _____
16. Missing leg: hyp = 17, leg = 8 → _____
17. Number of positive factors of 24 = _____
18. LCM of 6, 8, and 12 = _____
19. 19×21 = _____
20. 24×26 = _____

TEST C2D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Competition Number Sense, Algebra & Geometry

Date: _____ Time: _____ Score: _____ / 20

1. $1 + 2 + 3 + ... + 10$ = _____
2. $1 + 3 + 5 + 7 + 9 + 11$ = _____
3. 35^2 = _____
4. $25^2 - 15^2$ = _____
5. $24 \times (5/6 - 3/8)$ = _____
6. $1 + 1/(1 + 1/2)$ = _____
7. GCF(15, 20) × LCM(15, 20) = _____
8. Sum of positive factors of 12 = _____
9. $((35 - 7) \times 4) \div (14 + 2)$ = _____
10. $(12^2 + 5^2) \div 13$ = _____
11. Distance from (-3, 5) to (5, -10) = _____
12. 3D diagonal $\sqrt{(3^2 + 4^2 + 12^2)}$ = _____
13. Solve for x: $x + y = 12$, $x - y = 6$: _____
14. Minimum value of $y = x^2 - 6x + 14$: _____
15. Sum of roots of $2x^2 - 14x + 9 = 0$: _____
16. Hypotenuse for legs 7 and 24 = _____
17. $(45 \times 12) \div 18$ = _____
18. $1000 - 384$ = _____
19. $99 + 98 + 97$ = _____
20. $2/3 \times 3/4 \times 4/5 \times 5/6$ = _____

Section 9: Trigonometry, Vectors & Polar Coordinates

Special angles, right-triangle projections, inverse trig, and vector magnitudes. • Foundation (Test 9A) & Core Fluency (Test 9B)

TEST 9A • LEVEL 1: FOUNDATION Goal: 4 min

Exact Special-Angle Trig Ratios & Degree/Radian Conversions

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	$\sin(0°) =$	_____
2.	$\cos(0°) =$	_____
3.	$\sin(30°) =$	_____
4.	$\cos(60°) =$	_____
5.	$\sin(90°) =$	_____
6.	$\cos(90°) =$	_____
7.	$\tan(0°) =$	_____
8.	$\tan(45°) =$	_____
9.	$2 \times \sin(30°) =$	_____
10.	$4 \times \cos(60°) =$	_____
11.	$\sin(30°) + \cos(60°) =$	_____
12.	$5 \times \tan(45°) =$	_____
13.	Acute angle ϑ (deg): $\sin(\vartheta) = 0.5 \rightarrow$	_____
14.	Acute angle ϑ (deg): $\cos(\vartheta) = 0.5 \rightarrow$	_____
15.	Acute angle ϑ (deg): $\tan(\vartheta) = 1 \rightarrow$	_____
16.	Right triangle: $90° - 35° =$	_____
17.	Right triangle: $90° - 62° =$	_____
18.	$180°$ in radians (multiple of pi) =	_____
19.	$90°$ in radians =	_____
20.	$60°$ in radians =	_____

TEST 9B • LEVEL 2: CORE FLUENCY Goal: 5 min

Right-Triangle Projections & Vector Magnitudes

Date: _____ **Time:** _____ **Score:** _____ / 20

1.	Hyp = 10, $\vartheta = 30° \rightarrow$ Opp leg y =	_____
2.	Hyp = 12, $\vartheta = 60° \rightarrow$ Adj leg x =	_____
3.	Hyp = 20, $\vartheta = 30° \rightarrow$ Opp leg y =	_____
4.	Hyp = 16, $\vartheta = 60° \rightarrow$ Adj leg x =	_____
5.	Adj = 9, $\vartheta = 45° \rightarrow$ Opp leg y =	_____
6.	Opp = 7, $\vartheta = 30° \rightarrow$ Hypotenuse r =	_____
7.	Adj = 5, $\vartheta = 60° \rightarrow$ Hypotenuse r =	_____
8.	Polar ($r = 10, \vartheta = 0°$) $\rightarrow$ x =	_____
9.	Polar ($r = 10, \vartheta = 0°$) $\rightarrow$ y =	_____
10.	Polar ($r = 8, \vartheta = 90°$) $\rightarrow$ y =	_____
11.	Polar ($r = 14, \vartheta = 60°$) $\rightarrow$ x =	_____
12.	Polar ($r = 18, \vartheta = 30°$) $\rightarrow$ y =	_____
13.	Vector $\langle 3, 4 \rangle \rightarrow$ Magnitude r =	_____
14.	Vector $\langle 6, 8 \rangle \rightarrow$ Magnitude r =	_____
15.	Vector $\langle 5, 12 \rangle \rightarrow$ Magnitude r =	_____
16.	Vector $\langle 8, 6 \rangle \rightarrow$ Magnitude r =	_____
17.	Vector $\langle 9, 12 \rangle \rightarrow$ Magnitude r =	_____
18.	Vector $\langle 8, 15 \rangle \rightarrow$ Magnitude r =	_____
19.	Vector $\langle 12, 16 \rangle \rightarrow$ Magnitude r =	_____
20.	Vector $\langle 7, 24 \rangle \rightarrow$ Magnitude r =	_____

Section 9: Trigonometry, Vectors & Polar Coordinates

Special angles, right-triangle projections, inverse trig, and vector magnitudes. • Multi-Step Speed (Test 9C) & Advanced Challenge (Test 9D)

TEST 9C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Trigonometric Identities, Squares & Vector Operations

Date: _____ Time: _____ Score: _____ / 20

1. $\sin^2(30°) =$ _____
2. $\cos^2(30°) =$ _____
3. $\sin^2(45°) =$ _____
4. $\cos^2(45°) =$ _____
5. $\tan^2(60°) =$ _____
6. $\sin^2(37°) + \cos^2(37°) =$ _____
7. $5 \sin^2(20°) + 5 \cos^2(20°) =$ _____
8. $4 \cos^2(30°) =$ _____
9. $8 \sin^2(45°) =$ _____
10. $2 \sin(30°) \cos(60°) =$ _____
11. $\langle 2, 3\rangle + \langle 1, 1\rangle \rightarrow$ Magnitude r = _____
12. $\langle 4, 1\rangle + \langle 2, 7\rangle \rightarrow$ Magnitude r = _____
13. $\langle 8, 5\rangle - \langle 3, -7\rangle \rightarrow$ Magnitude r = _____
14. Dot product $\langle 2, 3\rangle \cdot \langle 4, 5\rangle =$ _____
15. Dot product $\langle 3, -2\rangle \cdot \langle 4, 6\rangle =$ _____
16. Area of d: a = 8, b = 10, C = 30° → _____
17. Area of d: a = 6, b = 12, C = 30° → _____
18. $12 \cos(60°) + 8 \sin(30°) =$ _____
19. $20 \sin(30°) - 6 \tan(45°) =$ _____
20. $\sec(60°) = 1 / \cos(60°) =$ _____

TEST 9D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

4-Sig-Fig Calculator Projections, Polar Angles & Law of Cosines

Date: _____ Time: _____ Score: _____ / 20

1. $12 \cos(45°) + 5 \sin(45°)$ (4 dec) = _____
2. $10 \cos(30°)$ (4 dec) = _____
3. $10 \sin(45°)$ (4 dec) = _____
4. $10 \tan(30°)$ (4 dec) = _____
5. Vector $\langle 3, 4\rangle$ angle ϑ (deg, 2 dec) = _____
6. Vector $\langle 4, 3\rangle$ angle ϑ (deg, 2 dec) = _____
7. Vector $\langle 5, 5\rangle$ angle ϑ (deg) = _____
8. Vector $\langle 0, -10\rangle$ angle ϑ (deg) = _____
9. Law of Cosines: a=3, b=5, C=60° → $c^2 =$ _____
10. Law of Cosines: a=8, b=5, C=60° → c = _____
11. Law of Sines: a=10, A=30°, B=90° → b = _____
12. $2 \sin(15°) \cos(15°) = \sin(30°) =$ _____
13. $\cos^2(15°) - \sin^2(15°) = \cos(30°)$ (4 dec) = _____
14. $1 + \tan^2(45°) = \sec^2(45°) =$ _____
15. $\sec^2(30°) - \tan^2(30°) =$ _____
16. max value of $3 \sin(x) + 4 \cos(x) =$ _____
17. max value of $5 \sin(x) + 12 \cos(x) =$ _____
18. max value of $8 \sin(x) - 15 \cos(x) =$ _____
19. Period of sin(2x) in degrees = _____
20. Period of cos(4x) in degrees = _____

Section 10: Exponents, Logarithms & Growth Models

Powers, fractional exponents, base-10 and natural logarithms, and exponential growth. • Foundation (Test 10A) & Core Fluency (Test 10B)

TEST 10A • LEVEL 1: FOUNDATION — Goal: 4 min

Integer Powers, Zero/Negative Exponents & Roots

Date: _____ **Time:** _____ **Score:** _____ / 20

1. $2^0 =$ _____
2. $e^0 =$ _____
3. $2^4 =$ _____
4. $2^5 =$ _____
5. $2^6 =$ _____
6. $2^8 =$ _____
7. $2^{10} =$ _____
8. $3^3 =$ _____
9. $3^4 =$ _____
10. $5^3 =$ _____
11. $5^4 =$ _____
12. $10^3 =$ _____
13. $10^4 =$ _____
14. 2^{-1} (decimal) $=$ _____
15. 2^{-2} (decimal) $=$ _____
16. 5^{-1} (decimal) $=$ _____
17. 10^{-2} (decimal) $=$ _____
18. $16^{1/2} =$ _____
19. $27^{1/3} =$ _____
20. $8^{2/3} =$ _____

TEST 10B • LEVEL 2: CORE FLUENCY — Goal: 5 min

Base-10 Logs, Binary Logs & Natural Logarithm Identities

Date: _____ **Time:** _____ **Score:** _____ / 20

1. $\log_{10}(1) =$ _____
2. $\log_{10}(10) =$ _____
3. $\log_{10}(100) =$ _____
4. $\log_{10}(1000) =$ _____
5. $\log_{10}(10000) =$ _____
6. $\log_{10}(0.1) =$ _____
7. $\log_{10}(0.01) =$ _____
8. $\ln(1) =$ _____
9. $\ln(e^1) =$ _____
10. $\ln(e^3) =$ _____
11. $\ln(e^{-2}) =$ _____
12. $e^{\ln 5} =$ _____
13. $e^{2 \ln 3} =$ _____
14. $10^{\log_{10} 7} =$ _____
15. $\log_2(8) =$ _____
16. $\log_2(32) =$ _____
17. $\log_2(64) =$ _____
18. $\log_3(27) =$ _____
19. $\log_5(125) =$ _____
20. $\log_{10}(2) + \log_{10}(5) =$ _____

Section 10: Exponents, Logarithms & Growth Models

Powers, fractional exponents, base-10 and natural logarithms, and exponential growth. • Multi-Step Speed (Test 10C) & Advanced Challenge (Test 10D)

TEST 10C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Logarithm Laws, Half-Life Decay & Compound Growth

Date: _____ **Time:** _____ **Score:** _____ / 20

1. $\log_{10}(4) + \log_{10}(25) =$ _____
2. $\log_{10}(20) + \log_{10}(50) =$ _____
3. $\log_{10}(200) - \log_{10}(2) =$ _____
4. $\log_{10}(5000) - \log_{10}(5) =$ _____
5. $\log_{2}(24) - \log_{2}(3) =$ _____
6. $\log_{3}(18) - \log_{3}(2) =$ _____
7. Solve: $2^{x} = 64$ _____
8. Solve: $3^{x} = 81$ _____
9. Solve: $5^{x - 1} = 25$ _____
10. Solve: $\log_{10}(x) = 3$ _____
11. Solve: $\log_{2}(x) = 5$ _____
12. Half-life (1 period): $100 \times (0.5)^{1} =$ _____
13. Half-life (2 periods): $100 \times (0.5)^{2} =$ _____
14. Half-life (3 periods): $100 \times (0.5)^{3} =$ _____
15. Half-life (4 periods): $160 \times (0.5)^{4} =$ _____
16. Doubling (3 periods): $50 \times 2^{3} =$ _____
17. Tripling (2 periods): $20 \times 3^{2} =$ _____
18. 1 year at 10%: $1000 \times 1.10 =$ _____
19. 2 years at 10%: $1000 \times (1.10)^{2} =$ _____
20. 2 years at 5%: $1000 \times (1.05)^{2} =$ _____

TEST 10D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Continuous Compounding, Doubling Time & Change of Base

Date: _____ **Time:** _____ **Score:** _____ / 20

1. Doubling time: $\ln(2) \div 0.07$ (4 dec) = _____
2. Doubling time: $\ln(2) \div 0.05$ (4 dec) = _____
3. Continuous: $500 \times e^{0.04 \times 5}$ (4 dec) = _____
4. Continuous: $1500 \times e^{0.05 \times 10}$ (2 dec) = _____
5. Monthly: $250 \times (1 + 0.06 \div 12)^{24}$ (2 dec) = _____
6. 3 years at 10%: $1000 \times (1.10)^{3} =$ _____
7. Change of base: $\ln(8) \div \ln(2) =$ _____
8. Change of base: $\ln(81) \div \ln(3) =$ _____
9. Change of base: $\ln(125) \div \ln(5) =$ _____
10. Change of base: $\log_{10}(32) \div \log_{10}(2) =$ _____
11. $\log_{4}(64) =$ _____
12. $\log_{8}(64) =$ _____
13. $\log_{4}(8)$ (decimal) = _____
14. $\log_{9}(27)$ (decimal) = _____
15. $27^{2/3} =$ _____
16. $16^{3/4} =$ _____
17. $32^{3/5} =$ _____
18. $64^{2/3} =$ _____
19. Solve: $4^{x} = 32$ (decimal) = _____
20. Solve: $9^{x} = 27$ (decimal) = _____

Section 11: Combinatorics, Probability & Statistics

Factorials, permutations, combinations, probability ratios, and dataset statistics. • Foundation (Test 11A) & Core Fluency (Test 11B)

TEST 11A • LEVEL 1: FOUNDATION Goal: 4 min

Factorials, Sample Means, Medians & Ranges

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **0! =** _____
2. **1! =** _____
3. **2! =** _____
4. **3! =** _____
5. **4! =** _____
6. **5! =** _____
7. **6! =** _____
8. **Sum: 1 + 2 + 3 + 4 + 5 =** _____
9. **Mean of {2, 4, 6, 8, 10} =** _____
10. **Mean of {10, 20, 30} =** _____
11. **Mean of {4, 7, 10, 11} =** _____
12. **Median of {3, 5, 9, 12, 15} =** _____
13. **Median of {2, 6, 10, 14} =** _____
14. **Range of {4, 9, 15, 22} =** _____
15. **Mode of {2, 5, 5, 8, 9} =** _____
16. **Fair coin P(Heads) (decimal) =** _____
17. **Fair 6-sided die P(roll = 6) =** _____
18. **Fair 6-sided die P(even) (decimal) =** _____
19. **Fair 6-sided die P(roll > 4) =** _____
20. **3 × 4 Choices (Fundamental Rule) =** _____

TEST 11B • LEVEL 2: CORE FLUENCY Goal: 5 min

Permutations nPr, Combinations nCr & Factorial Quotients

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **5! ÷ 4! =** _____
2. **6! ÷ 5! =** _____
3. **5P2 = 5! ÷ 3! =** _____
4. **6P2 = 6! ÷ 4! =** _____
5. **6P3 = 6! ÷ 3! =** _____
6. **7P2 = 7! ÷ 5! =** _____
7. **4C2 = 4! ÷ (2! × 2!) =** _____
8. **5C2 = 5! ÷ (2! × 3!) =** _____
9. **5C3 = 5! ÷ (3! × 2!) =** _____
10. **6C2 = 6! ÷ (2! × 4!) =** _____
11. **6C3 = 6! ÷ (3! × 3!) =** _____
12. **7C2 = 7! ÷ (2! × 5!) =** _____
13. **7C3 = 7! ÷ (3! × 4!) =** _____
14. **8C2 = 8! ÷ (2! × 6!) =** _____
15. **8C3 = 8! ÷ (3! × 5!) =** _____
16. **9C2 = 9! ÷ (2! × 7!) =** _____
17. **10C2 = 10! ÷ (2! × 8!) =** _____
18. **10C3 = 10! ÷ (3! × 7!) =** _____
19. **12C2 = 12! ÷ (2! × 10!) =** _____
20. **100C1 =** _____

Section 11: Combinatorics, Probability & Statistics

Factorials, permutations, combinations, probability ratios, and dataset statistics. • Multi-Step Speed (Test 11C) & Advanced Challenge (Test 11D)

TEST 11C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Independent Events, Arrangements & Weighted Averages

Date: _____ Time: _____ Score: _____ / 20

1. **2 coins: P(both Heads) (decimal) =** _____
2. **3 coins: P(all 3 Heads) (decimal) =** _____
3. **3 coins: P(at least 1 Tail) (decimal) =** _____
4. **2 dice: Total outcomes (6 × 6) =** _____
5. **2 dice: Ways to roll sum = 7 =** _____
6. **2 dice: P(sum = 7) =** _____
7. **2 dice: P(sum = 2) =** _____
8. **2 dice: P(sum = 12) =** _____
9. **Arrangements of letters in CAT (3!) =** _____
10. **Arrangements of MATH (4!) =** _____
11. **Arrangements of STACK (5!) =** _____
12. **Arrangements of MOM (3! ÷ 2!) =** _____
13. **Arrangements of BOOK (4! ÷ 2!) =** _____
14. **Arrangements of LEVEL (5! ÷ (2!2!)) =** _____
15. **Handshakes among 6 people (6C2) =** _____
16. **Handshakes among 8 people (8C2) =** _____
17. **Diagonals in a hexagon (6C2 - 6) =** _____
18. **Mean of 5 numbers is 12 → Sum =** _____
19. **Mean of {4, 8, 12, x} is 10 → x =** _____
20. **1 ÷ 6C2 (4 dec) =** _____

TEST 11D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Binomial Probability, Pascal's Triangle & Variance

Date: _____ Time: _____ Score: _____ / 20

1. **Row 4 sum: 4C0+4C1+4C2+4C3+4C4 =** _____
2. **Row 5 sum: 5C0 + ... + 5C5 = 2^5 =** _____
3. **Row 6 sum: 6C0 + ... + 6C6 = 2^6 =** _____
4. **4 coins: P(exactly 2 Heads) = 6/16 =** _____
5. **4 coins: P(exactly 3 Heads) = 4/16 =** _____
6. **5 coins: Ways to get 2 Heads (5C2) =** _____
7. **5 coins: P(exactly 2 Heads) = 10/32 =** _____
8. **Bag (3 Red, 2 Blue): P(2 Red) = 3C2/5C2 =** _____
9. **Bag (4 Red, 2 Blue): P(2 Red) = 4C2/6C2 =** _____
10. **Committee of 2 Boys (from 4) & 2 Girls (from 3) =** _____
11. **Circular arrangements of 5 people (4!) =** _____
12. **Circular arrangements of 6 people (5!) =** _____
13. **Variance of {2, 4, 6} (pop sd^2: ((4+0+4)/3)) =** _____
14. **Sample SD of {2, 5, 8} ($\sqrt{((9+0+9)/2)}$) =** _____
15. **Sample SD of {10, 14, 18} =** _____
16. **z-score: x = 85, mean = 75, sd = 5 →** _____
17. **z-score: x = 64, mean = 70, sd = 4 →** _____
18. **Expected value: 0.5(10) + 0.5(20) =** _____
19. **Expected value of 1 fair die (1..6) =** _____
20. **Expected value of 2 fair dice sum =** _____

Section 12: Base Conversions, Binary & Bitwise Logic

Binary, octal, and hexadecimal place value, bit shifting, and boolean gates. • Foundation (Test 12A) & Core Fluency (Test 12B)

TEST 12A • LEVEL 1: FOUNDATION Goal: 4 min

4-Bit Binary & Decimal Conversions

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **Binary 0001_2 in decimal =** _____
2. **Binary 0010_2 in decimal =** _____
3. **Binary 0011_2 in decimal =** _____
4. **Binary 0100_2 in decimal =** _____
5. **Binary 0101_2 in decimal =** _____
6. **Binary 0110_2 in decimal =** _____
7. **Binary 0111_2 in decimal =** _____
8. **Binary 1000_2 in decimal =** _____
9. **Binary 1001_2 in decimal =** _____
10. **Binary 1010_2 in decimal =** _____
11. **Binary 1100_2 in decimal =** _____
12. **Binary 1111_2 in decimal =** _____
13. **Decimal 5 in 4-bit binary =** _____
14. **Decimal 6 in 4-bit binary =** _____
15. **Decimal 9 in 4-bit binary =** _____
16. **Decimal 10 in 4-bit binary =** _____
17. **Decimal 12 in 4-bit binary =** _____
18. **Decimal 13 in 4-bit binary =** _____
19. **Decimal 14 in 4-bit binary =** _____
20. **Decimal 15 in 4-bit binary =** _____

TEST 12B • LEVEL 2: CORE FLUENCY Goal: 5 min

8-Bit Binary, Octal (Base 8) & Hexadecimal (Base 16)

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **Binary 10000_2 in decimal =** _____
2. **Binary 10100_2 in decimal =** _____
3. **Binary 11111_2 in decimal =** _____
4. **Binary 100000_2 in decimal =** _____
5. **Binary 1000000_2 in decimal =** _____
6. **Binary 11111111_2 in decimal =** _____
7. **Hex A_{16} in decimal =** _____
8. **Hex C_{16} in decimal =** _____
9. **Hex F_{16} in decimal =** _____
10. **Hex 10_{16} in decimal =** _____
11. **Hex $1A_{16}$ in decimal =** _____
12. **Hex 20_{16} in decimal =** _____
13. **Hex $2F_{16}$ in decimal =** _____
14. **Hex FF_{16} in decimal =** _____
15. **Decimal 16 in hexadecimal =** _____
16. **Decimal 31 in hexadecimal =** _____
17. **Decimal 64 in hexadecimal =** _____
18. **Decimal 255 in hexadecimal =** _____
19. **Decimal 8 in octal (Base 8) =** _____
20. **Decimal 64 in octal (Base 8) =** _____

Section 12: Base Conversions, Binary & Bitwise Logic

Binary, octal, and hexadecimal place value, bit shifting, and boolean gates. • Multi-Step Speed (Test 12C) & Advanced Challenge (Test 12D)

TEST 12C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Bitwise Boolean Logic (AND, OR, XOR) & Bit Shifts

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **1 AND 1 =** _____
2. **1 AND 0 =** _____
3. **1 OR 0 =** _____
4. **1 XOR 1 =** _____
5. **1 XOR 0 =** _____
6. **12 (1100_2) AND 10 (1010_2) =** _____
7. **12 (1100_2) OR 10 (1010_2) =** _____
8. **12 (1100_2) XOR 10 (1010_2) =** _____
9. **6 (0110_2) AND 3 (0011_2) =** _____
10. **6 (0110_2) OR 3 (0011_2) =** _____
11. **6 (0110_2) XOR 3 (0011_2) =** _____
12. **15 (1111_2) AND 9 (1001_2) =** _____
13. **15 (1111_2) XOR 9 (1001_2) =** _____
14. **255 AND 15 =** _____
15. **192 AND 240 =** _____
16. **Left shift: 5 << 1 (5 × 2) =** _____
17. **Left shift: 3 << 2 (3 × 4) =** _____
18. **Left shift: 7 << 3 (7 × 8) =** _____
19. **Right shift: 24 >> 2 (24 ÷ 4) =** _____
20. **Right shift: 64 >> 3 (64 ÷ 8) =** _____

TEST 12D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Mixed-Base Arithmetic, Two's Complement & Hex Masks

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **Binary sum: $1010_2 + 0101_2$ (in dec) =** _____
2. **Binary sum: $1100_2 + 0110_2$ (in dec) =** _____
3. **Hex sum: $A_{16} + 6_{16}$ (in Hex) =** _____
4. **Hex sum: $F_{16} + 1_{16}$ (in Hex) =** _____
5. **Hex sum: $1A_{16} + 16_{16}$ (in Hex) =** _____
6. **Octal 17_8 in decimal =** _____
7. **Octal 77_8 in decimal =** _____
8. **Octal 100_8 in hexadecimal =** _____
9. **Binary 10101100_2 in hexadecimal =** _____
10. **Binary 11110000_2 in hexadecimal =** _____
11. **Hex $B4_{16}$ in 8-bit binary =** _____
12. **Decimal 4095 ($2^{12} - 1$) in Hex =** _____
13. **Max unsigned 8-bit integer ($2^8 - 1$) =** _____
14. **Max unsigned 10-bit integer ($2^{10} - 1$) =** _____
15. **4-bit bitwise NOT of 5 (15 - 5) =** _____
16. **8-bit bitwise NOT of 200 (255 - 200) =** _____
17. **8-bit Two's Complement of -1 (in Hex) =** _____
18. **8-bit Two's Complement of -2 (in Hex) =** _____
19. **(255 AND 240) >> 4 =** _____
20. **(170 XOR 85) (10101010_2 ^ 01010101_2) =** _____

Cumulative Review C3: All-Level Championship (Sections 1–12)

Grand championship speed tests spanning arithmetic, algebra, trig, logs, stats, and CS. • Foundation (Test C3A) & Core Fluency (Test C3B)

TEST C3A • LEVEL 1: FOUNDATION Goal: 5 min

Core Cross-Topic Speed: Arithmetic, Fractions, Powers & Binary

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **(18 - 6) × 3 =** _____
2. **3/4 + 1/2 (decimal) =** _____
3. **GCF of 18 and 24 =** _____
4. **LCM of 6 and 8 =** _____
5. **Solve: 3x - 5 = 16** _____
6. **Hypotenuse for legs 5 and 12 =** _____
7. **sin(30°) + cos(60°) =** _____
8. **tan(45°) × 8 =** _____
9. **2^5 =** _____
10. **$\log_{10}(1000)$ =** _____
11. **$\ln(e^4)$ =** _____
12. **5! =** _____
13. **5C2 =** _____
14. **Mean of {4, 8, 12, 16} =** _____
15. **Binary 1010_2 in decimal =** _____
16. **Decimal 15 in 4-bit binary =** _____
17. **Hex 10_{16} in decimal =** _____
18. **12 AND 10 =** _____
19. **15^2 =** _____
20. **1000 - 375 =** _____

TEST C3B • LEVEL 2: CORE FLUENCY Goal: 5 min

Intermediate Championship: Quadratics, Vectors, Logs & Combinatorics

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **Sum of double 6 and near-double (8 + 9) =** _____
2. **((35 - 7) × 4) ÷ (14 + 2) =** _____
3. **(1/3 + 1/6) × 18 =** _____
4. **Greater root of $x^2 + 5x + 6 = 0$:** _____
5. **Positive root of $x^2 - 25 = 0$:** _____
6. **Slope from (-2, 1) to (2, 13) =** _____
7. **Vector <8, 15> → Magnitude r =** _____
8. **Hyp = 20, $\vartheta = 30°$ → Opp leg y =** _____
9. **2^{10} =** _____
10. **$\log_2(64)$ =** _____
11. **$1000 \times (1.05)^2$ =** _____
12. **Half-life (3 periods): $100 \times (0.5)^3$ =** _____
13. **6P3 =** _____
14. **7C3 =** _____
15. **8C3 =** _____
16. **Decimal 255 in hexadecimal =** _____
17. **Decimal 64 in octal (Base 8) =** _____
18. **12 OR 10 =** _____
19. **12 XOR 10 =** _____
20. **192 AND 240 =** _____

Cumulative Review C3: All-Level Championship (Sections 1–12)

Grand championship speed tests spanning arithmetic, algebra, trig, logs, stats, and CS. • Multi-Step Speed (Test C3C) & Advanced Challenge (Test C3D)

TEST C3C • LEVEL 3: MULTI-STEP SPEED Goal: 6 min

Advanced Championship: Multi-Step Rationals, Trig & Growth

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **$(3\ 1/3 \div 1\ 1/4) + 2/5$ (4 dec) =** _____
2. **$1 \div (1/4 + 1/6)$ (decimal) =** _____
3. **Distance from (-3, 5) to (5, -10) =** _____
4. **Evaluate $2x^2 + 4x - 5$ at $x = 3$:** _____
5. **Sum of roots of $x^2 - 11x + 28 = 0$:** _____
6. **$12 \cos(45°) + 5 \sin(45°)$ (4 dec) =** _____
7. **$\sin^2(25°) + \cos^2(25°)$ =** _____
8. **Dot product $\langle 3, 4\rangle \cdot \langle 5, -2\rangle$ =** _____
9. **Doubling time: $\ln(2) \div 0.07$ (4 dec) =** _____
10. **Continuous: $500 \times e^{0.04 \times 5}$ (4 dec) =** _____
11. **Continuous: $1500 \times e^{0.05 \times 10}$ (2 dec) =** _____
12. **Monthly: $250 \times (1 + 0.06 \div 12)^{24}$ (2 dec) =** _____
13. **10C3 =** _____
14. **Arrangements of LEVEL (5! ÷ 4) =** _____
15. **2 dice: P(sum = 7) =** _____
16. **Decimal 4095 ($2^{12} - 1$) in Hex =** _____
17. **Hex $2F_{16}$ in decimal =** _____
18. **Left shift: $9 << 3$ (9×8) =** _____
19. **24×26 =** _____
20. **35^2 =** _____

TEST C3D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Grand Invitational Speed Challenge

Date: _____ **Time:** _____ **Score:** _____ / 20

1. **$1 + 2 + 3 + \ldots + 20$ =** _____
2. **$99 + 98 + 97 + 96 + 95$ =** _____
3. **$25^2 - 15^2$ =** _____
4. **$24 \times (5/6 - 3/8)$ =** _____
5. **GCF(24, 36) × LCM(24, 36) =** _____
6. **Number of positive factors of 36 =** _____
7. **3D diagonal $\sqrt{(3^2 + 4^2 + 12^2)}$ =** _____
8. **Minimum value of $y = x^2 - 8x + 19$:** _____
9. **Solve for x: $2x + y = 15$, $x - y = 3$:** _____
10. **Law of Cosines: a=8, b=5, C=60° → c =** _____
11. **max value of $5 \sin(x) + 12 \cos(x)$ =** _____
12. **Change of base: $\ln(125) \div \ln(5)$ =** _____
13. **$64^{2/3}$ =** _____
14. **Solve: $4^x = 32$ (decimal) =** _____
15. **Row 6 sum: 6C0 + 6C1 + ... + 6C6 =** _____
16. **5 coins: P(exactly 2 Heads) (decimal) =** _____
17. **Sample SD of {10, 14, 18} =** _____
18. **Binary 10101100_2 in hexadecimal =** _____
19. **8-bit Two's Complement of -1 (in Hex) =** _____
20. **(170 XOR 85) =** _____

Official Answer Key — Sections 1 to 3

Complete verified answers for all four progressive difficulty levels (Tests A, B, C & D, problems #1–#20).

Section 1: Step Counting, Benchmarks & Sequences (Foundational to Advanced)

Test 1A (Level 1: Foundation)		Test 1B (Level 2: Core Fluency)		Test 1C (Level 3: Multi-Step Speed)		Test 1D (Level 4: Advanced Challenge)	
1. 4	11. 6	1. 6	11. 15	1. 36	11. 100	1. 15	11. 150
2. 6	12. 12	2. 8	12. 20	2. 48	12. 90	2. 30	12. 36
3. 9	13. 13	3. 10	13. 30	3. 43	13. 245	3. 25	13. 60
4. 5	14. 14	4. 15	14. 30	4. 51	14. 165	4. 30	14. 50
5. 9	15. 14	5. 30	15. 50	5. 70	15. 200	5. 40	15. 250
6. 6	16. 13	6. 16	16. 30	6. 55	16. 255	6. 75	16. 90
7. 9	17. 12	7. 14	17. 22	7. 125	17. 300	7. 28	17. 70
8. 7	18. 10	8. 25	18. 60	8. 80	18. 300	8. 52	18. 350
9. 4	19. 16	9. 15	19. 55	9. 135	19. 480	9. 90	19. 55
10. 7	20. 16	10. 35	20. 100	10. 100	20. 300	10. 56	20. 110

Section 2: Number Bonds & Missing Addends (Foundational to Advanced)

Test 2A (Level 1: Foundation)		Test 2B (Level 2: Core Fluency)		Test 2C (Level 3: Multi-Step Speed)		Test 2D (Level 4: Advanced Challenge)	
1. 7	11. 3	1. 14	11. 8	1. 39	11. 33	1. 190	11. 40
2. 7	12. 11	2. 13	12. 9	2. 59	12. 43	2. 155	12. 35
3. 8	13. 11	3. 15	13. 9	3. 64	13. 34	3. 65	13. 185
4. 8	14. 11	4. 15	14. 9	4. 63	14. 56	4. 82	14. 320
5. 9	15. 12	5. 17	15. 11	5. 33	15. 83	5. 175	15. 200
6. 9	16. 12	6. 7	16. 7	6. 36	16. 95	6. 135	16. 275
7. 9	17. 12	7. 6	17. 14	7. 37	17. 37	7. 129	17. 265
8. 4	18. 6	8. 8	18. 7	8. 35	18. 52	8. 70	18. 180
9. 3	19. 5	9. 7	19. 15	9. 22	19. 45	9. 100	19. 275
10. 5	20. 4	10. 9	20. 9	10. 26	20. 46	10. 30	20. 400

Section 3: Complements & Make-10 Bridging (Foundational to Advanced)

Test 3A (Level 1: Foundation)		Test 3B (Level 2: Core Fluency)		Test 3C (Level 3: Multi-Step Speed)		Test 3D (Level 4: Advanced Challenge)	
1. 3	11. 19	1. 17	11. 34	1. 70	11. 66	1. 600	11. 57.5
2. 6	12. 13	2. 15	12. 37	2. 55	12. 84	2. 350	12. 146
3. 2	13. 15	3. 13	13. 42	3. 38	13. 67	3. 725	13. 163
4. 7	14. 14	4. 14	14. 24	4. 72	14. 93	4. 616	14. 283
5. 1	15. 12	5. 13	15. 32	5. 26	15. 64	5. 258	15. 455
6. 4	16. 13	6. 16	16. 36	6. 81	16. 63	6. 481	16. 735
7. 8	17. 15	7. 8	17. 23	7. 17	17. 66	7. 137	17. 297
8. 5	18. 12	8. 23	18. 55	8. 43	18. 125	8. 903	18. 291
9. 14	19. 14	9. 26	19. 67	9. 45	19. 139	9. 6.6	19. 600
10. 17	20. 16	10. 23	20. 47	10. 67	20. 136	10. 3.25	20. 3355

Official Answer Key — Sections 4 to 5

Complete verified answers for all four progressive difficulty levels (Tests A, B, C & D, problems #1–#20).

Section 4: Doubles, Equal Groups & Squares (Foundational to Advanced)

Test 4A (Level 1: Foundation)		Test 4B (Level 2: Core Fluency)		Test 4C (Level 3: Multi-Step Speed)		Test 4D (Level 4: Advanced Challenge)	
1. 6	11. 13	1. 26	11. 12	1. 54	11. 96	1. 121	11. 99
2. 8	12. 15	2. 28	12. 20	2. 76	12. 120	2. 144	12. 143
3. 10	13. 17	3. 30	13. 18	3. 92	13. 200	3. 169	13. 224
4. 12	14. 19	4. 32	14. 30	4. 130	14. 37	4. 196	14. 399
5. 14	15. 22	5. 36	15. 32	5. 156	15. 48	5. 225	15. 624
6. 16	16. 24	6. 50	16. 42	6. 60	16. 65	6. 256	16. 899
7. 18	17. 21	7. 70	17. 12	7. 72	17. 21	7. 400	17. 450
8. 20	18. 23	8. 90	18. 18	8. 92	18. 30	8. 625	18. 244
9. 9	19. 12	9. 31	19. 25	9. 100	19. 78	9. 900	19. 200
10. 11	20. 18	10. 49	20. 24	10. 140	20. 50	10. 1225	20. 400

Cumulative Review C1: Number Sense (Sections 1–4) (Foundational to Advanced)

Test C1A (Level 1: Foundation)		Test C1B (Level 2: Core Fluency)		Test C1C (Level 3: Multi-Step Speed)		Test C1D (Level 4: Advanced Challenge)	
1. 8	11. 13	1. 27	11. 24	1. 63	11. 23	1. 21	11. 5.65
2. 10	12. 15	2. 20	12. 40	2. 32	12. 56	2. 42	12. 36.2
3. 9	13. 7	3. 9	13. 16	3. 84	13. 65	3. 535	13. 390
4. 5	14. 8	4. 8	14. 23	4. 36	14. 100	4. 218	14. 260
5. 2	15. 15	5. 16	15. 36	5. 76	15. 144	5. 446	15. 240
6. 7	16. 18	6. 32	16. 57	6. 55	16. 225	6. 563	16. 280
7. 16	17. 15	7. 32	17. 35	7. 96	17. 90	7. 196	17. 105
8. 14	18. 9	8. 48	18. 65	8. 76	18. 40	8. 625	18. 475
9. 16	19. 20	9. 35	19. 36	9. 120	19. 165	9. 399	19. 650
10. 13	20. 15	10. 21	20. 55	10. 55	20. 67	10. 899	20. 49

Section 5: Multi-Step Operations & Grouping (Grades 3–8+)

Test 5A (Level 1: Foundation)		Test 5B (Level 2: Core Fluency)		Test 5C (Level 3: Multi-Step Speed)		Test 5D (Level 4: Advanced Challenge)	
1. 24	11. 18	1. 36	11. 8	1. 69	11. 24	1. 110	11. 15
2. 35	12. 8	2. 5	12. 40	2. 28	12. 32	2. 108	12. 32
3. 48	13. 27	3. 5	13. 30	3. 4	13. 17	3. 8	13. 31
4. 63	14. 22	4. 5	14. 25	4. 10	14. 120	4. 10	14. 24
5. 8	15. 8	5. 36	15. 35	5. 72	15. 6	5. 100	15. 90
6. 9	16. 10	6. 8	16. 10	6. 72	16. 60	6. 10	16. 30
7. 9	17. 20	7. 40	17. 40	7. 90	17. 60	7. 100	17. 120
8. 7	18. 20	8. 90	18. 48	8. 28	18. 5	8. 96	18. 16
9. 18	19. 30	9. 80	19. 5	9. 8	19. 10	9. 600	19. 10
10. 5	20. 5	10. 10	20. 72	10. 56	20. 120	10. 5	20. 10

Official Answer Key — Sections 6 to 8

Complete verified answers for all four progressive difficulty levels (Tests A, B, C & D, problems #1–#20).

Section 6: Fractions, Decimals & Reciprocals (Grades 4–9+)

Test 6A (Level 1: Foundation)		Test 6B (Level 2: Core Fluency)		Test 6C (Level 3: Multi-Step Speed)		Test 6D (Level 4: Advanced Challenge)	
1. 3/4	11. 0.25	1. 1/2	11. 1/2	1. 3.75	11. 10	1. 1	11. 2/5
2. 2/5	12. 0.75	2. 1/2	12. 12	2. 2.25	12. 0.875	2. 2	12. 1/5
3. 5/8	13. 0.2	3. 5/8	13. 4/3	3. 2.25	13. 0.125	3. 2	13. 2
4. 3/10	14. 0.6	4. 1/2	14. 1.25	4. 1.75	14. 0.375	4. 2.4	14. 2
5. 3/4	15. 2/5	5. 5/6	15. 2.5	5. 6	15. 0.625	5. 1	15. 16
6. 1/4	16. 1/2	6. 1/6	16. 2	6. 15	16. 0.875	6. 10	16. 5/3
7. 1	17. 8	7. 7/12	17. 3	7. 10	17. 3	7. 14	17. 7/4
8. 3/4	18. 5	8. 5/12	18. 1.5	8. 10	18. 3	8. 1	18. 2.5
9. 5/8	19. 9	9. 3/8	19. 1	9. 6	19. 1.5	9. 17	19. 11
10. 0.5	20. 10	10. 1/2	20. 12	10. 6	20. 10	10. 18	20. 19

Section 7: Primes, GCF/LCM & Nested Expressions (Grades 5–10+)

Test 7A (Level 1: Foundation)		Test 7B (Level 2: Core Fluency)		Test 7C (Level 3: Multi-Step Speed)		Test 7D (Level 4: Advanced Challenge)	
1. 12	11. 125	1. 12	11. 7	1. 50	11. 50	1. 216	11. 28
2. 18	12. 2	2. 15	12. 4	2. 30	12. 5	2. 300	12. 10
3. 20	13. 17	3. 16	13. 6	3. 20	13. 40	3. 36	13. 10
4. 24	14. 4	4. 14	14. 10	4. 12	14. 24	4. 24	14. 40
5. 30	15. 5	5. 24	15. 72	5. 40	15. 4	5. 36	15. 7
6. 36	16. 6	6. 36	16. 60	6. 5	16. 10	6. 120	16. 4
7. 8	17. 12	7. 60	17. 200	7. 70	17. 60	7. 6	17. 5
8. 16	18. 12	8. 70	18. 25	8. 8	18. 8	8. 8	18. 10
9. 27	19. 24	9. 20	19. 16	9. 36	19. 5	9. 9	19. 60
10. 64	20. 10	10. 10	20. 36	10. 30	20. 125	10. 10	20. 13

Section 8: Algebra, Quadratics & Coordinate Geometry (Grades 6–10+)

Test 8A (Level 1: Foundation)		Test 8B (Level 2: Core Fluency)		Test 8C (Level 3: Multi-Step Speed)		Test 8D (Level 4: Advanced Challenge)	
1. 17	11. 6	1. 3	11. 13	1. 20	11. 3	1. 9	11. 7
2. 13	12. 7	2. 2	12. 17	2. 25	12. 5	2. 20	12. 3
3. 18	13. 5	3. 3	13. 15	3. 4	13. 3	3. -11	13. 5
4. 12	14. 5	4. 3	14. 20	4. 17	14. 4	4. 24	14. 7
5. 14	15. 12	5. -2	15. 8	5. -2	15. 5	5. 5	15. 5
6. 12	16. 7	6. -0.5	16. 12	6. -3	16. -5	6. 3.5	16. 25
7. 23	17. -8	7. 5	17. 5	7. -3	17. 4	7. 16	17. 41
8. 9	18. 24	8. 7	18. 5	8. -3	18. -0.5	8. 0	18. 13
9. 35	19. -4	9. 5	19. 10	9. 3	19. 3	9. 4	19. 7
10. 6	20. 25	10. 10	20. 17	10. 2	20. 5	10. 3	20. 9

Official Answer Key — Sections C2 to 10

Complete verified answers for all four progressive difficulty levels (Tests A, B, C & D, problems #1–#20).

Cumulative Review C2: Intermediate Math (Sections 1–8) (Grades 5–10+)

Test C2A (Level 1: Foundation)		Test C2B (Level 2: Core Fluency)		Test C2C (Level 3: Multi-Step Speed)		Test C2D (Level 4: Advanced Challenge)	
1. 27	11. 12	1. 1.75	11. 3	1. 0.875	11. -5	1. 55	11. 17
2. 64	12. 20	2. 3.75	12. 20	2. 2.4	12. 7	2. 36	12. 13
3. 36	13. 10	3. 1.25	13. 10	3. 14	13. 12	3. 1225	13. 9
4. 5	14. 18	4. 1.5	14. 3	4. 17	14. 7	4. 400	14. 5
5. 0.75	15. 14	5. 15	15. 4	5. 12	15. 12	5. 11	15. 7
6. 0.25	16. 7	6. 10	16. -2	6. 40	16. 15	6. 5/3	16. 25
7. 12	17. 14	7. 32	17. 2	7. 70	17. 8	7. 300	17. 30
8. 18	18. 5	8. 50	18. 28	8. 25	18. 24	8. 28	18. 616
9. 24	19. 10	9. 12	19. 225	9. 3	19. 399	9. 7	19. 294
10. 6	20. 4	10. 24	20. 625	10. -0.5	20. 624	10. 13	20. 1/3

Section 9: Trigonometry, Vectors & Polar Coordinates (Grades 8–12+)

Test 9A (Level 1: Foundation)		Test 9B (Level 2: Core Fluency)		Test 9C (Level 3: Multi-Step Speed)		Test 9D (Level 4: Advanced Challenge)	
1. 0	11. 1	1. 5	11. 7	1. 0.25	11. 5	1. 12.0208	11. 20
2. 1	12. 5	2. 6	12. 9	2. 0.75	12. 10	2. 8.6603	12. 0.5
3. 0.5	13. 30	3. 10	13. 5	3. 0.5	13. 13	3. 7.0711	13. 0.8660
4. 0.5	14. 60	4. 8	14. 10	4. 0.5	14. 23	4. 5.7735	14. 2
5. 1	15. 45	5. 9	15. 13	5. 3	15. 0	5. 53.13	15. 1
6. 0	16. 55	6. 14	16. 10	6. 1	16. 20	6. 36.87	16. 5
7. 0	17. 28	7. 10	17. 15	7. 5	17. 18	7. 45	17. 13
8. 1	18. pi	8. 10	18. 17	8. 3	18. 10	8. -90	18. 17
9. 1	19. pi/2	9. 0	19. 20	9. 4	19. 4	9. 19	19. 180
10. 2	20. pi/3	10. 8	20. 25	10. 0.25	20. 2	10. 7	20. 90

Section 10: Exponents, Logarithms & Growth Models (Grades 8–12+)

Test 10A (Level 1: Foundation)		Test 10B (Level 2: Core Fluency)		Test 10C (Level 3: Multi-Step Speed)		Test 10D (Level 4: Advanced Challenge)	
1. 1	11. 625	1. 0	11. -2	1. 2	11. 32	1. 9.9021	11. 3
2. 1	12. 1000	2. 1	12. 5	2. 3	12. 50	2. 13.8629	12. 2
3. 16	13. 10000	3. 2	13. 9	3. 2	13. 25	3. 610.7014	13. 1.5
4. 32	14. 0.5	4. 3	14. 7	4. 3	14. 12.5	4. 2473.08	14. 1.5
5. 64	15. 0.25	5. 4	15. 3	5. 3	15. 10	5. 281.79	15. 9
6. 256	16. 0.2	6. -1	16. 5	6. 2	16. 400	6. 1331	16. 8
7. 1024	17. 0.01	7. -2	17. 6	7. 6	17. 180	7. 3	17. 8
8. 27	18. 4	8. 0	18. 3	8. 4	18. 1100	8. 4	18. 16
9. 81	19. 3	9. 1	19. 3	9. 3	19. 1210	9. 3	19. 2.5
10. 125	20. 4	10. 3	20. 1	10. 1000	20. 1102.5	10. 5	20. 1.5

Official Answer Key — Sections 11 to C3

Complete verified answers for all four progressive difficulty levels (Tests A, B, C & D, problems #1–#20).

Section 11: Combinatorics, Probability & Statistics (Grades 7–12+)

Test 11A (Level 1: Foundation)		Test 11B (Level 2: Core Fluency)		Test 11C (Level 3: Multi-Step Speed)		Test 11D (Level 4: Advanced Challenge)	
1. 1	11. 8	1. 5	11. 20	1. 0.25	11. 120	1. 16	11. 24
2. 1	12. 9	2. 6	12. 21	2. 0.125	12. 3	2. 32	12. 120
3. 2	13. 8	3. 20	13. 35	3. 0.875	13. 12	3. 64	13. 8/3
4. 6	14. 18	4. 30	14. 28	4. 36	14. 30	4. 0.375	14. 3
5. 24	15. 5	5. 120	15. 56	5. 6	15. 15	5. 0.25	15. 4
6. 120	16. 0.5	6. 42	16. 36	6. 1/6	16. 28	6. 10	16. 2
7. 720	17. 1/6	7. 6	17. 45	7. 1/36	17. 9	7. 0.3125	17. -1.5
8. 15	18. 0.5	8. 10	18. 120	8. 1/36	18. 60	8. 0.3	18. 15
9. 6	19. 1/3	9. 10	19. 66	9. 6	19. 16	9. 0.4	19. 3.5
10. 20	20. 12	10. 15	20. 100	10. 24	20. 0.0667	10. 18	20. 7

Section 12: Base Conversions, Binary & Bitwise Logic (Grades 7–12+)

Test 12A (Level 1: Foundation)		Test 12B (Level 2: Core Fluency)		Test 12C (Level 3: Multi-Step Speed)		Test 12D (Level 4: Advanced Challenge)	
1. 1	11. 12	1. 16	11. 26	1. 1	11. 5	1. 15	11. 10110100
2. 2	12. 15	2. 20	12. 32	2. 0	12. 9	2. 18	12. FFF
3. 3	13. 0101	3. 31	13. 47	3. 1	13. 6	3. 10	13. 255
4. 4	14. 0110	4. 32	14. 255	4. 0	14. 15	4. 10	14. 1023
5. 5	15. 1001	5. 64	15. 10	5. 1	15. 192	5. 30	15. 10
6. 6	16. 1010	6. 255	16. 1F	6. 8	16. 10	6. 15	16. 55
7. 7	17. 1100	7. 10	17. 40	7. 14	17. 12	7. 63	17. FF
8. 8	18. 1101	8. 12	18. FF	8. 6	18. 56	8. 40	18. FE
9. 9	19. 1110	9. 15	19. 10	9. 2	19. 6	9. AC	19. 15
10. 10	20. 1111	10. 16	20. 100	10. 7	20. 8	10. F0	20. 255

Cumulative Review C3: All-Level Championship (Sections 1–12) (Grades 7–12+)

Test C3A (Level 1: Foundation)		Test C3B (Level 2: Core Fluency)		Test C3C (Level 3: Multi-Step Speed)		Test C3D (Level 4: Advanced Challenge)	
1. 36	11. 4	1. 29	11. 1102.5	1. 3.0667	11. 2473.08	1. 210	11. 13
2. 1.25	12. 120	2. 7	12. 12.5	2. 2.4	12. 281.79	2. 485	12. 3
3. 6	13. 10	3. 9	13. 120	3. 17	13. 120	3. 400	13. 16
4. 24	14. 10	4. -2	14. 35	4. 25	14. 30	4. 11	14. 2.5
5. 7	15. 10	5. 5	15. 56	5. 11	15. 1/6	5. 864	15. 64
6. 13	16. 1111	6. 3	16. FF	6. 12.0208	16. FFF	6. 9	16. 0.3125
7. 1	17. 16	7. 17	17. 100	7. 1	17. 47	7. 13	17. 4
8. 8	18. 8	8. 10	18. 14	8. 7	18. 72	8. 3	18. AC
9. 32	19. 225	9. 1024	19. 6	9. 9.9021	19. 624	9. 6	19. FF
10. 3	20. 625	10. 6	20. 192	10. 610.7014	20. 1225	10. 7	20. 255

Inside-Out Math & Stack Reference

Five inside-out mathematical patterns and their matching verification sequences on your four-line calculator.

---

1. START AT THE DEEPEST INNER EXPRESSION (ZERO PARENTHESES)

Evaluate the most deeply nested quantity first so intermediate results wait in Register Y. Example: For 100 ÷ (3 + 4×5), compute 4×5 + 3 = 23 first, then type 100 and swap:

Keycap Sequence: 

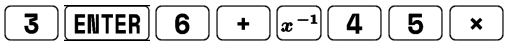
2. INSTANT DOUBLING & SQUARING ([ENTER] [+] AND [ENTER] [×])

Pressing [ENTER] duplicates Register X into Register Y without retyping digits. To double 18 and add 5 in 5 keystrokes:

Keycap Sequence: 

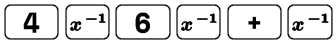
3. NON-COMMUTATIVE SWAP (X SWAP Y) & RECIPROCAL ([1/x])

If you computed the denominator D first and then typed numerator N into X, press [x swap y] [÷]—or invert D in place with [1/x] and multiply:

Keycap Sequence: 

4. PARALLEL RESISTANCE & HARMONIC SUMS (AUTOMATIC STACK LIFT)

To compute 1 / (1/4 + 1/6), never use memory registers—automatic stack lift pushes the first reciprocal into Y while the second is computed:

Keycap Sequence: 

5. VECTOR MAGNITUDE & POLAR CONVERSION (TO POLAR)

To find the hypotenuse or magnitude of components (x=5, y=12) in a single operation on the SC-32:

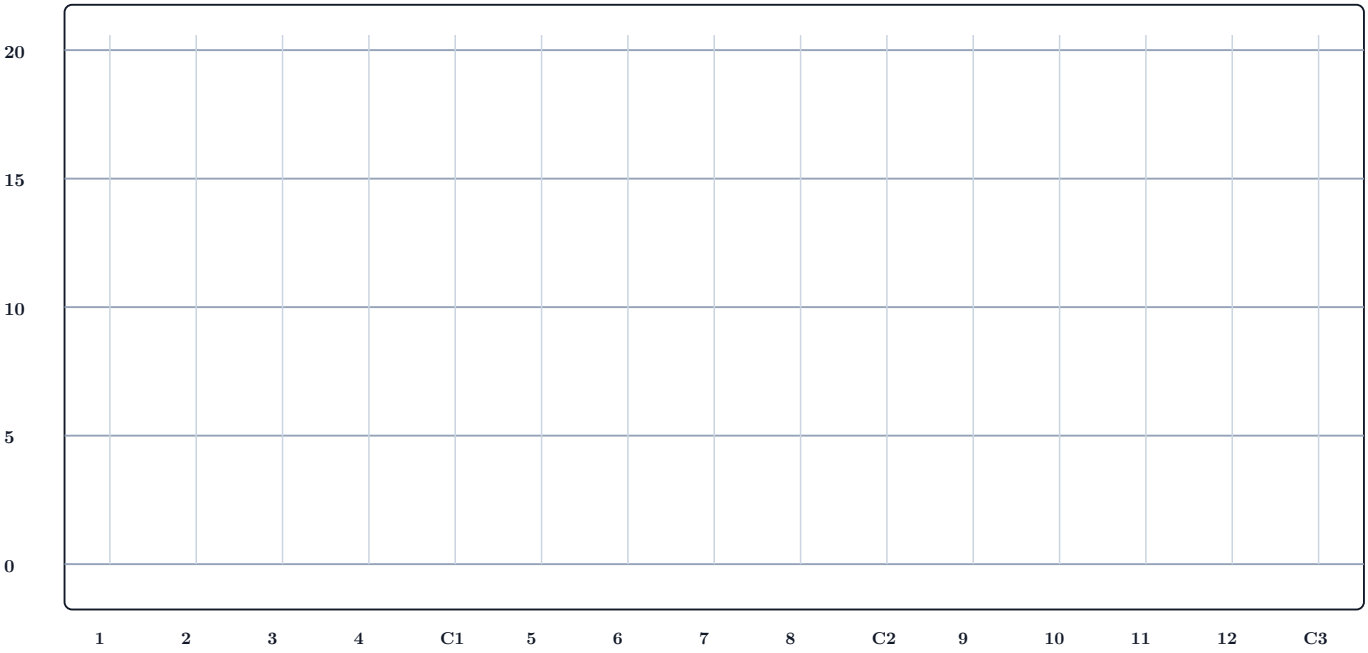
Keycap Sequence: 

Speed & Fluency Progress Graph & Certificate

Plot your highest score (out of 20) across all 15 sections and record your level completion.

---

1. 15-SECTION SCORE PROGRESSION GRAPH (PLOT YOUR SCORE FROM 0 TO 20)



CERTIFICATE OF MATHEMATICAL SPEED & FLUENCY

This certifies that: _____

has completed the progressive practice test series in Stack Speed & Fluency, demonstrating mathematical accuracy, multi-step structural fluency, and confident verification across Levels 1 through 4.

[ ] Level 1: Foundation Mastery (Tests A) **[ ] Level 2: Core Fluency Mastery (Tests B)**

[ ] Level 3: Multi-Step Speed (Tests C) **[ ] Level 4: Advanced Challenge (Tests D)**

Teacher / Parent Signature: _____ Date: _____